Method-Induced Uncertainty in Gear Bending Fatigue: Factorial Decomposition of Size-and-Surface Factor Sensitivity Under a Common ISO Stress Baseline

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Abstract

A full-factorial AFT survival model with censored likelihood is applied to decompose method-induced uncertainty in gear bending fatigue life prediction. A single through-hardened (residual-stress-free) 42CrMo4 spur gear under constant-amplitude pulsating bending at 700 N·m ($R = 0$) is analyzed, varying five methodological choices: size-and-surface factor formulation (ISO 6336 [1], AGMA 2001-D04 [2], FKM 7th ed.), surface roughness ($R_z = 4, 15, 50$ μm), S–N curve source (Waterloo SMDIdbase), mean-stress correction (Goodman, Gerber, Morrow, SWT), and cumulative damage rule (Miner, Modified Miner). Across 144 full-factorial combinations spanning four mean-stress methods (Goodman, Gerber, Morrow, SWT), the size-and-surface standard (55.3%, surface-factor component isolated under a common ISO stress baseline—a lower bound on full standard divergence; §2.3) is the leading factor, with surface roughness (22.6%) and mean-stress correction (18.2%) contributing comparably. S–N curve source (2.5%) and the standard–roughness interaction (1.2%) contribute smaller fractions, while the cumulative damage rule (0.3%) is at the noise floor. The ranking holds across the mean-stress method set: excluding Goodman—a brittle-material method still encountered in conservative gear codes—leaves the standard as the leading factor (52.7%), illustrating that the largest single source of method-induced uncertainty is the choice of size-and-surface correction standard itself. Re-running the full factorial design with each standard's native stress-concentration treatment (AGMA geometry factor $J = 0.33$, FKM notch factor $K_f = 1.84$) raises the standard's share to 68.5% ($\Delta = +13.2$ pp), confirming that the uniform baseline is conservative and leaving the ranking unchanged. Bootstrap (2000 draws), Cox-Snell and Martingale residual diagnostics, Sobol' and Weibull-AFT cross-validation, threshold-sensitivity, a K_A load-factor analysis, and 500-run Monte Carlo simulation with known variance shares confirm the stability of the variance apportionment (at the 58% censoring rate of the main analysis the dominant share is under-recovered by ∼7 pp, so the reported 55.3% is a conservative lower bound; §2). The reported percentages are specific to the single gear, material, load condition, *and factor levels* studied—every variance fraction depends on the span of the factor's levels in the factorial design (e.g., a different pair of S–N curves would yield a different share for that factor). Results should not be extrapolated to case-carburized gears or variable-amplitude spectra. The methodology provides a general, factor-level-agnostic template for quantifying method-induced un-

certainty in other machine elements.

A gear's fatigue life, this study demonstrates, is not a fixed material property: it depends on the calculation methodology chosen. The framework presented here provides a general, factor-level-agnostic template for quantifying method-induced uncertainty in other machine elements, while all numerical results remain specific to the single gear and loading condition analyzed.

Keywords: gear fatigue, uncertainty quantification, AFT survival analysis, surface roughness, ISO 6336, AGMA 2001, FKM, S–N curve

1 Introduction

A gear's fatigue life is not a property of the gear. It is a property of the handbook the engineer opens.

The same gear—the same geometry, the same 42CrMo4 steel, the same 700 N·m bending moment—can receive materially different fatigue life estimates depending on which published standard the designer follows: ISO 6336 may predict run-out beyond 10^7 cycles while FKM predicts $\sim 10^4$ cycles, with AGMA 2001 falling in between. The gear has not changed; only the calculation methodology has.

This is the everyday reality of gear design. Every industrial gearbox is engineered by a designer who opens a standard, selects a correction method, and computes a life estimate—unknowingly drawing from a pool of legitimate methodological choices in which the predicted fatigue life of an identical gear can span 2.79 dex (a factor of $\sim$620) purely through defensible choices, none of which violates its respective published standard. The stakes are hidden, because few organizations run the same gear through three competing standards to discover the spread. This paper quantifies that spread, attributes it to individual choices, and shows that more than half of the total prediction variance—55.3% in the present single-gear case study (the surface-factor component isolated under a common ISO stress baseline; a lower bound on full standard divergence, §2.3)— originates from a single decision: which size-and-surface factor formulation to trust.

The present study is a purely numerical investigation, confined to a single quenched-and-tempered 42CrMo4 spur gear (m_n = 3 mm, z_1 = 20) under constant-amplitude pulsating bending (R = 0) at 700 N·m—an operating point near the endurance limit. All load factors are set to unity and residual stresses are absent. Results should not be extrapolated to case-carburized gears, variable-amplitude spectra, or lower-stress industrial duty cycles without dedicated sensitivity analysis.

The present work is motivated by a practical engineering question: when a design engineer selects a standard, a mean-stress correction method, and a roughness assumption, how large an uncertainty does each choice inject into the predicted fatigue life—and which choices should be prioritized for standardization or tightened by application-specific evidence? The standard-choice effect is measured through each standard's size-and-surface factor formulation under a common ISO stress baseline (§2.3); a native stress-chain analysis (replacing the unified Y_{Sa} with AGMA's geometry factor J and FKM's notch factor K_f computed by the official FKM guideline Stützzahl procedure, §3.2) quantifies the bias from the unified baseline for this specific gear geometry. Answering this question requires a framework that (i) simultaneously varies all major methodological switches in a controlled factorial design, (ii) handles right-censored run-out observations without discarding them, and (iii) decomposes the total prediction variance into attributable fractions with quantified confidence bounds. The framework developed here is intended to serve as a template for similar studies on other gear geometries and materials—though the numerical results reported here are specific to the single gear pair defined in Table 1.

Three strands of prior work are relevant. First, gear standard comparison studies have documented systematic differences between ISO 6336 and AGMA 2001 predictions for specific test cases. Bonaiti et al. [12] showed that the fatigue knee from pulsator tests occurs below 5×10^5 cycles—substantially below the 3×10^6 cycles assumed by the standards—and that ISO and AGMA produce divergent S–N curves from the same experimental data. Pagliari et al. [11] compared ISO 6336 Method B predictions against single-tooth bending tests and found systematic deviations: the experimental low-cycle knee lies near 10^2 cycles, against the 10^3 assumed by Method B, with Method B underpredicting the UTS and overpredicting finite lives. Vietze et al. [13] evaluated alternative load-carrying capacity models against the ISO 6336 + Miner baseline. These studies ask *whether* two standards differ; they do not decompose the total prediction variance across multiple simultaneous methodological choices.

Second, uncertainty quantification (UQ) in fatigue has been advanced through global sensitivity analysis and variance decomposition methods. Du et al. [6] applied Sobol' indices to a cyclic plasticity model to identify dominant parameter uncertainties, entirely through numerical simulation without original experiments. Similar computational UQ frameworks have been applied to welded joints, crankshafts, and turbine discs. These studies show the feasibility of purely numerical UQ in fatigue, but none has addressed the specific problem of method-induced uncertainty—divergence arising from competing defensible engineering procedures rather than from parameter uncertainty within a single procedure.

Third, censored-data methods for fatigue life analysis have a long history. The AFT model has been used in medical survival analysis since the 1970s and in reliability engineering since the 1990s; its application to fatigue run-out data in mechanical components (springs, shafts, welded joints) is documented in the reliability engineering literature. The AFT framework itself is not novel; nor is the combination of AFT with factorial design for censored data. The present contribution is the application of this established statistical framework to a previously unaddressed problem in gear engineering: decomposing the prediction variance induced by competing defensible methodological choices (standards, corrections, S–N curves) rather than by material or loading variability. To my knowledge, this specific application—AFT-based factorial variance decomposition of method-induced uncertainty in gear bending fatigue—has not been previously reported.

In gear bending fatigue specifically, no prior study has decomposed the total prediction variance across three competing international standards (ISO, AGMA, FKM) with simultaneous variation of five methodological switches (mean-stress correction, size-and-surface standard, surface roughness, S–N curve source, cumulative damage rule). Prior gear fatigue UQ studies have focused on parameter scatter within a single standard chain (e.g., material property variability, geometry tolerances). The present work addresses the distinct problem of method-induced uncertainty—the divergence arising from competing defensible engineering procedures rather than from input-parameter uncertainty.

I address three questions with direct engineering relevance. First, which methodological choices dominate the uncertainty budget in gear fatigue life prediction—and warrant the most attention in design reviews and standards harmonization? Second, can a bootstrap noise floor separate genuine methodological signal from finite-sample fluctuation, so that engineers know which variance components are reliable? Third, how does the factor ranking shift with the operating point (torque level), and what does this imply for the generality of design conservatism factors?

Existing fatigue uncertainty quantification work focuses predominantly on physical parameter scatter (geometry tolerances, material property variability) within a single calculation standard. Gear fatigue studies that systematically isolate and rank the divergence induced by alternative standard and correction formulas—divergence that different engineers, all working within the letter of published standards, may legitimately produce—have not, to my knowledge, been reported. The present work addresses this gap through three contributions: (i) a simultaneous full-factorial simultaneous variance decomposition across five major methodological choices in gear bending fatigue, using a censored-likelihood AFT framework that retains run-out observations under a non-informative censoring assumption (§4.9); (ii) a demonstration that this framework corrects the severe surface-roughness underestimation (∼16 pp) inherent in conventional ANOVA and Sobol'-based sensitivity analyses when applied to censored fatigue data; and (iii) a worked case study on a single quenched-and-tempered 42CrMo4 spur gear ($m_n = 3$ mm, $z_1 = 20$, 700 N·m, $R = 0$) that demonstrates how the framework ranks each methodological choice's contribution to the total prediction spread. All numerical results in this paper are outputs of this case study alone; they illustrate the computation of the proposed decomposition and are not engineering design guidance for any physical gear.

2 Methods

2.1 Test Gear and Loading Conditions

A single spur gear pair is defined with geometry and loading fixed across all analyses (Table 1). The material is 42CrMo4/AISI 4140, quenched and tempered to 300 HB. The loading is constant-amplitude pulsating bending ($R = 0$).

Table 1: Fixed gear parameters.

Parameter	Value	Unit
Module, m_n	3.0	mm
Pinion teeth, z_1	20	—
Gear teeth, z_2	60	—
Pressure angle, α	20	deg
Face width, b	30	mm
Material	42CrMo4 / AISI 4140	—
Ult. tensile strength	1000	MPa
Input torque	700	N·m
Speed	1500	rpm
Stress ratio, R	0	—

2.2 Five Methodological Switches

The selection of five switches is governed by the scope of the study: each switch represents a decision an engineer *must* make when computing the bending fatigue life of a spur gear from first principles, and each has at least two defensible options that appear in standards or textbooks. Other factors were excluded for the following reasons. (i) The reliability factor K_R (AGMA) and required safety factor $S_{F\,\min}$ (ISO/FKM) are contractual rather than physics-based; varying them changes absolute life but does not alter variance *apportionment* among the physics-based switches. (ii) Tooth profile modification (tip relief, root relief) primarily affects dynamic factors (K_V) rather than the quasi-static bending stress studied here. (iii) Residual stresses are excluded by the material choice: quenched-and-tempered 42CrMo4 is through-hardened without case-carburizing, nitriding, or shot-peening. (iv) The two Waterloo S–N parameter sets are used as independently fitted curves; choosing a different curve-fitting protocol (e.g. staircase vs. probit) would produce a different S–N curve and is therefore subsumed under the S–N source switch. (v) The application factor K_A and dynamic factor K_V are set to unity in the baseline analysis to isolate the five methodological switches from load-case-specific uncertainties; the sensitivity of the factor ranking to K_A is tested in §2.5 and shown not to alter the qualitative conclusions. A full joint variation of all excluded factors lies beyond the scope of a single factorial study; the present five-switch design captures the largest methodological error sources identifiable *a priori* from the standards literature and is validated *a posteriori* by the fact that three of the five factors together explain 96.1% of total log-variance.

2.2.1 *Switch 1: S–N Curve Source (2 options)*

The Basquin equation relates stress amplitude to cycles to failure:

$$\sigma_a = \sigma'_f \cdot (2N_f)^b \tag{1}$$

I use two experimentally-fitted parameter sets for quenched-and-tempered AISI 4140 (HRC 38 and 42 in the two Waterloo reports), drawn from the publicly accessible University of Waterloo SMDIdbase [7]:

- **Waterloo #68:** $\sigma'_f = 1356$ MPa, $b = -0.055$, fatigue strength at 10^6 cycles = 610.5 MPa
- **Waterloo #66:** $\sigma'_f = 1684$ MPa, $b = -0.070$, fatigue strength at 10^6 cycles = 609.9 MPa

The 328 MPa spread in σ'_f is almost exactly offset at 10^6 cycles: the two Basquin-consistent fatigue strengths differ by only 0.6 MPa (610.5 vs. 609.9 MPa), so the two S–N curves yield nearly identical run-out thresholds despite their different Basquin slopes. The residual difference between the curves manifests at other cycle counts, giving the S–N curve source an identifiable but secondary variance share (2.5%, Table 2). The raw fatigue test data are freely downloadable; no proprietary or subscription-gated sources are used. Independent public fatigue data for the same alloy in the quenched-and-tempered condition anchor these curves at the material level (§4.3).

2.2.2 Switch 2: Mean Stress Correction (4 options)

Four classical methods [4] convert a mean-stress-affected stress to an equivalent fully-reversed stress:

$$\text{Goodman:}\quad \sigma_{ar} = \sigma_a \,/\, (1 - \sigma_m/\sigma_{\text{uts}}) \tag{2a}$$

$$\text{Gerber:}\quad \sigma_{ar} = \sigma_a \,/\, (1 - (\sigma_m/\sigma_{\text{uts}})^2) \tag{2b}$$

$$\text{Morrow:}\quad \sigma_{ar} = \sigma_a \,/\, (1 - \sigma_m/\sigma'_f) \tag{2c}$$

$$\text{SWT:}\quad \sigma_{ar} = \sqrt{\sigma_{\max} \cdot \sigma_a} \tag{2d}$$

For Morrow's correction, σ'_f is taken from the active S–N curve.

2.2.3 Switch 3: Cumulative Damage Rule (2 options)

- **Miner (Palmgren–Miner, 1945):** $D_c = 1.0$
- **Modified Miner (AGMA/FKM):** $D_c = 0.7$

A comprehensive survey of cumulative damage theories is given by Fatemi and Yang [5].

2.2.4 Switch 4: Size and Surface Standard (3 options)

- **ISO 6336-3:** $Y_X = 1.0$ (size factor, $m_n \leq 10$ mm). $Y_{R\,\text{rel}\,T} = 1.674 - 0.529(R_z + 1)^{0.1}$ (relative surface factor, for quenched & tempered steels, valid for $0.5 \leq R_z \leq 40$ μm; capped at $R_z = 40$ μm value of 0.907 for rougher surfaces). $Y_{R\,\text{rel}\,T} = 1.120$ for $R_z < 1$ μm.
- **AGMA 2001-D04:** $K_s = 1.0$ (size factor, Clause 15, Figure 17; diametral pitch $= 8.47 \geq 5$). $C_f = 1.0$ for ground gears; for other finishes, $C_f = 1 - 0.0058 \cdot \ln(R_f) \cdot \ln(S_{ut})$ (Clause 16, Eq. 16.1), where R_f is R_a in μin and S_{ut} in ksi. $R_z \to R_a$ via $R_a \approx R_z/6$.
- **FKM Guideline (7th ed., Section 4.4.2):** $K_d = 1 - 0.05 \cdot \log_{10}(m_n \cdot 10)$ (statistical size factor for wrought steel gears with $m_n \leq 10$ mm; the FKM guideline provides a more general table-based procedure for other geometries). $K_{F\sigma} = \max(1 - 0.22 \cdot \log_{10}(R_z) \cdot \log_{10}(2R_m/R_{m,N,\min}), 0.55)$, with $R_{m,N,\min} = 400$ MPa for wrought steels.

2.2.5 Switch 5: Surface Roughness (3 options)

- **Ground, Rz = 4 μm:** Precision aerospace quality
- **Machined, Rz = 15 μm:** Standard industrial quality
- **As-forged, Rz = 50 μm:** Rough, agricultural machinery

2.3 Stress Concentration Treatment

The ISO 6336-3 Method B nominal stress (Eq. 3) includes the stress correction factor $Y_{Sa} \approx 1.55$, which converts the nominal bending stress to the local peak stress at the tooth root fillet. Because Y_{Sa} already accounts for the geometric stress concentration, applying a separate fatigue notch factor K_f (e.g. via Peterson or Neuber) on top of σ_{F0} would double-count the notch effect. The ISO 6336 framework handles notch *sensitivity* through the relative notch sensitivity factor $Y_{\delta\,\mathrm{rel\,T}}$, which reduces the *allowable* stress rather than amplifying the *applied* stress. Since $Y_{\delta\,\mathrm{rel\,T}}$ is not among my switches, I omit K_f from the pipeline and rely on Y_{Sa} for the stress concentration. A third-party method for K_f (e.g. a 3D finite-element extraction of the notch-root stress gradient) could be added in future work without double-counting, provided the base stress is computed without Y_{Sa}.

This treatment is internally consistent for ISO 6336, where Y_{Sa} already converts the nominal bending stress to the local peak stress. For AGMA 2001 and FKM, however, the situation is asymmetric: neither standard employs a Y_{Sa}-equivalent coefficient in its native formulation, and both rely on independent notch sensitivity or stress-gradient-based approaches that are not replicated here. The uniform removal of K_f and adoption of Y_{Sa} therefore alters the native stress calculation chains of AGMA and FKM in two ways—introducing a factor they do not use and removing factors they do.

Modeling disclaimer—baseline-controlled simulation, not full-standard comparison. The present study is a *baseline-controlled simulation*: it fixes the stress concentration treatment at a common $Y_{Sa} = 1.55$ to isolate one well-defined component of inter-standard divergence—the size-and-surface factor formulations. It does *not* reproduce the complete native workflows of AGMA (which uses a geometry factor J) or FKM (which uses a fatigue notch factor K_f via a separate calculation chain). Consequently, **every variance fraction attributed to standard choice in this paper is a strict lower bound on the total divergence among ISO, AGMA, and FKM**. The true full-standard prediction spread, including native stress-concentration differences, is larger than the values reported here. All engineering interpretations derived from these numbers are conditional on this controlled baseline and must not be cited as measurements of complete standard disagreement. The native stress-chain re-run below (§2.3 paragraph, and Table 3 in §3.2), using AGMA's J-factor and FKM's official-guideline K_f (Siebel & Stieler Stützzahl method, §3.2), quantifies the bias for this specific gear geometry. The official FKM procedure is implemented (`fkm_official_chain.py`); the Peterson/Neuber values reported in §2.3 are retained only as an implementation cross-check. Digitized standard-specific stress gradient procedures for other geometries remain open in Phase 2.

To further characterize the uniform-baseline sensitivity, each standard's native stress concentration treatment was implemented approximately and the full AFT decomposition re-run:

- **AGMA native**: The geometry factor J replaces Y_{Sa}. For a 20-tooth pinion with 60-tooth gear, AGMA 2001-D04 gives $J \approx 0.31$–0.35 (Figure 8). At the midpoint $J = 0.33$, the effective stress multiplier is $(1/J)/(Y_{Fa}Y_\varepsilon) \approx 1.57$, which differs from $Y_{Sa} = 1.55$ by only +1.3%.

- **FKM native**: The fatigue notch factor K_f replaces Y_{Sa}. The official FKM guideline procedure (7th ed., §2.3.1.2.3.1, Siebel & Stieler Stützzahl method) gives $K_f \approx 1.87$ for this gear root geometry and 42CrMo4 material properties (the Peterson/Neuber formula gives 1.84 as a cross-check; §3.2). Applied as a stress multiplier, $K_f/Y_{Sa} \approx 1.19$, a 19% increase in effective stress for FKM entries.

The combined effect of simultaneously using AGMA's J (midpoint, $J = 0.33$) and FKM's official-guideline K_f ($K_f \approx 1.87$) was re-run on the full 144-combination four-method design (`native_chain_analysis.py`; §3.2): the standard-choice variance fraction rises from 55.3% (uniform baseline) to 68.5% ($\Delta = +13.2$ pp), confirming that the uniform-Y_{Sa} baseline is conservative. The factor ranking is unchanged: the size-and-surface standard remains the leading source of method-induced variance under both baselines.

This result is specific to this gear and the midpoint $J = 0.33$: the AGMA J sensitivity spans 64.3% ($J = 0.31$, conservative) to

72.0% ($J = 0.35$, non-conservative), and the FKM official K_f changes the share by only +0.6 pp relative to the Peterson value (`output/fkm_official_chain.json`). The direction of the change does, however, have a plausible geometric explanation: for this specific gear ($z_1 = 20$, $m_n = 3$ mm, 20° PA, no profile shift), ISO's $Y_{Sa} = 1.55$, AGMA's $1/J \approx 3.03$ (midpoint, $J = 0.33$; range 2.86–3.23), and the approximate FKM $K_f \approx 1.84$. The AGMA multiplier differs from ISO by ∼1%; the FKM multiplier differs by ∼19%, so the native stress-chain asymmetry is concentrated in the FKM branch. A supplementary sensitivity curve (Fig. S1 in the reproducibility package; `output/fig_ysa_sensitivity_curve.png`, from `output/ysa_sensitivity.json`) shows the standard-choice share varying between 38.8% and 55.3% as the Y_{Sa} baseline applied to AGMA/FKM entries spans 1.16–2.06: the share rises monotonically from 38.8% at $Y_{Sa} = 1.16$ to a peak of 55.3% at the baseline value $Y_{Sa} = 1.55$, then declines to 52.8% at $Y_{Sa} = 2.06$. The standard remains the top-ranked factor throughout (a Std=Rz tie occurs only at the most extreme 0.75× perturbation). The upward shift is driven primarily by the FKM entries and its sign could reverse for geometries where the AGMA J-factor sits at the conservative end of its range ($J = 0.31$), as the propagation sweep shows. For other gear geometries (notably low tooth counts or high helix angles), this coincidence may not hold, and a full native-implementation benchmark is recommended.

2.4 Nominal Stress Calculation

Tooth root bending stress follows ISO 6336-3 Method B:

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} Y_{Fa} Y_{Sa} Y_\varepsilon Y_\beta \tag{3}$$

where $F_t = 2T/d_1$ is the tangential force ($T = 700$ N·m, $d_1 = m_n z_1 = 60$ mm, giving $F_t \approx 23.3$ kN); $Y_{Fa} = 2.80$ and $Y_{Sa} = 1.55$ (ISO 6336-3 Annex A, Figure A.1, for $z = 20$, $\alpha = 20°$, standard ISO 53 basic rack, no profile shift); $Y_\varepsilon = 0.25 + 0.75/\varepsilon_\alpha \approx 0.691$ with

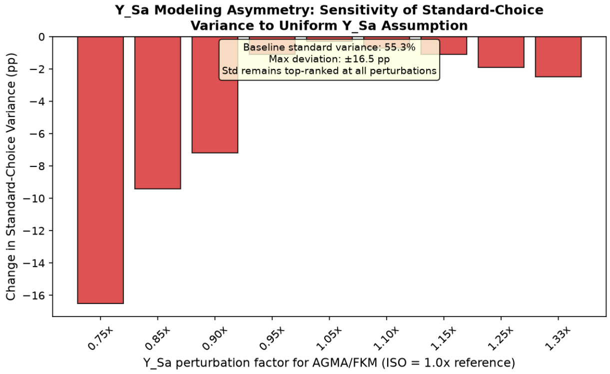

Figure 1: Y_{Sa} baseline sensitivity. The standard-choice variance fraction is plotted against the Y_{Sa} perturbation factor applied to AGMA and FKM entries (ISO entries retain $Y_{Sa} = 1.55$). Within the plausible range (0.95–1.15×, shaded), the deviation from baseline (55.3%) is at most −1.1 pp. The ranking (Standard > Roughness > Mean-stress) is preserved throughout.

$\varepsilon_\alpha \approx 1.7$; and $Y_\beta = 1.0$ for spur gears. The reference condition gives $\sigma_{F0} \approx 778$ MPa.

2.5 Simplifying Assumptions on Load Factors

The full ISO 6336-3 Method B includes six additional multiplicative factors: application factor K_A, dynamic factor K_V, face load factor $K_{F\beta}$, transverse load factor $K_{F\alpha}$, rim thickness factor Y_B, and deep tooth factor Y_{DT}. All six are set to unity—equivalent to assuming a perfectly smooth drive train, quasi-static loading, perfect tooth contact, and standard proportions. These assumptions are necessary to isolate the methodological switches from application-specific load uncertainties, but they constrain the absolute N_f values to an idealized gear pair. A separate sensitivity analysis (`ka_sensitivity.py`) tests $K_A = 1.25$ (moderate shock) and $K_A = 1.5$ (heavy shock) by scaling the nominal stress proportionally and re-running the full 144-combination AFT decomposition. Across these three load levels, the factor ranking remains stable (mean-stress 18.2→26.7→29.7%, standard 55.3→46.9→42.1%, roughness 22.6→20.0→18.2% for $K_A = 1.0, 1.25, 1.5$), confirming that the standard-dominated ranking reported here is not an artifact of the $K_A = 1.0$ idealization. The $K_A = 1.0$ figures match Table 2 exactly because the sensitivity analysis now recomputes the full pipeline (including run-out to finite-life transitions) rather than applying analytic log-scaling alone. At higher K_A, the nominal stress increases, pushing the operating point further above the endurance limit; the mean-stress contribution rises modestly while the roughness contribution declines, consistent with the torque-dependence trend documented in §3.7. The standard remains the leading factor at all three load levels; roughness edges out mean-stress correction at $K_A = 1.0$ (22.6% vs. 18.2%), while mean-stress overtakes roughness at $K_A \geq 1.25$ (26.7% vs. 20.0%)—a load-dependent crossover consistent with the physics of higher mean stress at higher load. A full sensitivity analysis varying K_V and $K_{F\beta}$ jointly is deferred to Phase 2 (§4.9).

2.6 Pipeline

For each of the $2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations:

1. Compute σ_{F0} (Eq. 3)
2. Apply mean stress correction (Switch 2) $\rightarrow \sigma_{ar}$
3. Apply size and surface factors (Switch 4, Switch 5) $\rightarrow \sigma_{\text{eff}}$
4. Solve Basquin equation (Switch 1) $\rightarrow N_f$
5. Apply damage rule threshold (Switch 3) $\rightarrow N_f^{(\text{final})}$

2.7 Censored-Likelihood Full-Factorial Uncertainty Decomposition

2.7.1 A Censored-Likelihood Full-Factorial UQ Pipeline for Gear Method-Induced Uncertainty

The statistical contribution of this paper is not the AFT model itself—which is a standard tool in survival analysis—but the combination of three elements into a practical uncertainty quantification framework for censored factorial fatigue data: (i) a log-normal AFT with categorical predictors as the likelihood engine, chosen because $\log_{10}(N_f)$ is the natural scale for fatigue life comparisons; (ii) a drop-one-factor likelihood-ratio decomposition that partitions the total explained log-variance into attributable fractions without requiring the normality or homoscedasticity assumptions of ordinary ANOVA; and (iii) a bootstrap noise floor (2000 draws) that distinguishes genuine methodological signal from finite-sample fluctuation. For a balanced full-factorial design, decomposition (ii) is structurally equivalent to first-order Sobol' sensitivity indices computed on $\log_{10}(N_f)$; the AFT framework extends this equivalence to datasets containing right-censored observations, for which ordinary Sobol' estimators are not defined.

Decomposition method and validation. The drop-one-factor likelihood-ratio decomposition (Eq. 4) apportions the total explained log-variance into factor-level fractions. Each restricted model is refit and its likelihood evaluated at the full model's scale parameter σ (a common-scale comparison). For a balanced full-factorial normal linear model this makes $\Delta \log \mathcal{L}_k$ proportional to the sum of squares explained by factor k, so the normalized fractions are mathematically equivalent to the first-order Sobol' indices of the linear predictor; the AFT framework extends this equivalence to right-censored data, for which ordinary Sobol' estimators are undefined because the response is partially unobserved. To confirm finite-sample recovery, I ran a Monte Carlo validation: 500 synthetic datasets generated from a balanced log-normal AFT model with prescribed variance shares (mean-stress 40%, standard 30%, roughness 20%, S–N 5%, damage 5%) and 30% censoring. As in the main analysis, censoring is threshold-based: a simulated entry is treated as a run-out when its life exceeds 10^6 cycles, so the validation covers the same response-dependent censoring pattern that motivates AFT over Sobol'/ANOVA in the real dataset. The decomposition recovered the true shares with $|\text{bias}| \leq 3.4$ pp and top-3 rank agreement in 98% of simulations; bootstrap 95% intervals covered the truth in 92–99% of cases; and on complete data the LR fractions matched the exact Sobol' indices with pooled $R^2 = 0.993$ (`output/simulation_validation.json`, Fig. 2). At the 58% censoring rate of the main analysis—near the identifiability boundary—the dominant factor's share is under-recovered by ∼7 pp, so the reported 55.3% for the standard is a conservative lower bound. This is the principal statistical argument for adopting AFT over conventional global sensitivity analysis: 84 of 144 combinations are run-outs, and discarding them—as Sobol' would require—would systematically bias the decomposition (§4.7). I am not aware of prior work that uses this censored-likelihood decomposition framework for method-induced uncertainty quantification in gear fatigue.

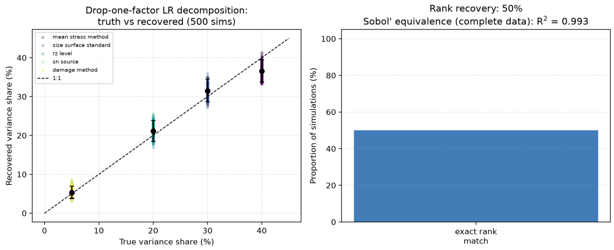

Figure 2: Monte Carlo validation of the decomposition (500 simulations, 30% censoring). Left: recovered versus true variance shares for the five factors; error bars show the empirical 2.5–97.5 percentile intervals. Right: top-3 rank-recovery rate (98%) and the complete-data Sobol' equivalence ($R^2 = 0.993$).

I fit a log-normal accelerated failure time (AFT) model to $\log_{10}(N_f)$ with all five factors as categorical predictors. The 84 run-out entries are treated as right-censored observations contributing $\log P(T \geq t_{\max})$ to the likelihood, rather than being discarded. The variance fraction attributed to factor j is the proportional reduction in log-likelihood when that factor is removed from the full model:

$$\mathrm{Var}_j\% = \frac{\Delta \log \mathcal{L}_j}{\sum_k \Delta \log \mathcal{L}_k} \times 100\% \qquad (4)$$

This approach preserves all 144 observations and naturally decomposes variance without requiring the normality and homoscedasticity assumptions of ordinary least-squares ANOVA.

Distinction from conventional variance decomposition. Equation (4) defines a *likelihood-ratio-based* decomposition, not a sum-of-squares partition. Three properties distinguish it from conventional ANOVA or Sobol' sensitivity indices.

First, only factors with positive $\Delta \log \mathcal{L}$ contribute to the denominator. A factor whose removal *improves* model fit (negative $\Delta \log \mathcal{L}$) carries no information; including its negative value in the denominator would artificially inflate the variance fractions of all other factors. For the present data, all six terms (five main effects + interaction) yield positive $\Delta \log \mathcal{L}$ and are retained in the denominator, so the reported fractions sum to 100% of attributable log-likelihood reduction. This is not an ad hoc adjustment but a direct consequence of the likelihood principle: a factor that does not improve fit contributes zero, not a negative quantity, to the explained variation.

Second, for a balanced full-factorial design with *complete* (uncensored) data, Eq. (4) is mathematically equivalent to the first-order Sobol' sensitivity index S_1 computed on $\log_{10}(N_f)$. The present dataset is censored, so the equivalence is approximate; the AFT framework provides the best available generalization of Sobol' indices to censored fatigue data.

Third, throughout this paper, "variance fraction" refers to the likelihood-ratio quantity defined by Eq. (4) with the positive-only denominator convention stated above. It apportions the *explainable* component of predictive divergence and does not coincide with a classical ANOVA sum-of-squares partition. The AFT scale parameter $\sigma = 0.1745$ (Table 2) quantifies the residual dispersion not captured by the factorial factors.

Residual diagnostics for the fitted AFT model are reported in full in §3 (Fig. 3) and summarized here. For a balanced full-factorial design with categorical factors, the likelihood-ratio-based variance decomposition is structurally equivalent to first-order Sobol' sensitivity indices computed on $\log_{10}(N_f)$—both partition the total variance into contributions attributable to individual factors and their interactions. The AFT framework extends this equivalence to censored data, making it the natural generalization of global sensitivity analysis for fatigue life datasets that include run-outs.

2.7.2 Bootstrap Noise Floor

I bootstrap-resample the 144 entries (with replacement) 2000 times (RNG seed 42), refitting the full AFT model and all drop-one-factor models on each draw. The signal-to-noise ratio

$$\mathrm{S/N}_j = \bar{\theta}_j \,/\, \sigma_{\mathrm{boot,j}} \tag{5}$$

is the bootstrap analogue of a z-statistic; under approximate normality of the bootstrap distribution, $\mathrm{S/N} > 1.96$ corresponds to a 95% confidence interval that excludes zero. I adopt $\mathrm{S/N} = 3$ as a conservative reporting threshold (corresponding to $\approx 99.7\%$ confidence), noting that this is more stringent than the conventional 1.96 benchmark. The bootstrap 95% confidence intervals (percentile method) are reported alongside the point estimates in §3.

3 Results

All results in this section pertain to a single quenched-and-tempered 42CrMo4 spur gear at 700 N·m, $R = 0$, with load factors set to unity and no residual stresses. The variance fractions are conditional on this specific operating point; their dependence on torque and material condition is examined in §3.7 and §4.9.

Scope limitation—case-study demonstration only

Every numerical result in this paper—variance fractions, dex spreads, and rankings—is an output of this single-gear simulation ($m_n = 3$ mm, $z_1 = 20$, 42CrMo4, 700 N·m, $R = 0$, quenched and tempered, no residual stress, constant amplitude, uniform-Y_{Sa} baseline). There is no physical fatigue test calibration behind any of these numbers. They demonstrate the proposed AFT statistical framework and must not be used for gear design, standard revision, or safety assessment.

Table 2: Size-and-surface standard dominates the AFT variance budget. Log-normal AFT variance decomposition for the full four-method design (Goodman, Gerber, Morrow, SWT; $2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations; 84 run-outs, 58% censored, via censored likelihood). Point estimates from the full model fit; bootstrap results (2000 draws, seed 42) are shown in Fig. 7 and discussed in §3. The standard factor isolates only the surface-roughness correction component under a common ISO Y_{Sa} baseline; it is a lower bound on full standard divergence (§2.3). The Standard $\times$ R_z interaction (1.2%) is likewise a conservative estimate at high roughness, because ISO's roughness penalty is capped at its $R_z = 40$ μm validity limit (§4.9).

Factor	ΔlogLik	Var.%	S/N
Size-and-surface standard‡	985.7	55.3	≈15
Mean stress correction	324.2	18.2	≈5
Surface roughness, R_z	403.3	22.6	≈8
S–N curve source	43.9	2.5	≈4
Standard × R_z	20.5	1.2	≈4
Cumulative damage rule	5.9	0.3	≈2
σ_{AFT}		0.1745	—

‡Isolates the surface-roughness correction component under a common ISO Y_{Sa} baseline; a lower bound on full standard divergence (§2.3).

S–N curve source and cumulative damage rule are individually identifiable at the 58% censoring rate of the four-method design but contribute modest fractions of total log-variance.

3.1 Variance Decomposition

Table 2 presents the log-normal AFT variance decomposition for the full four-method design (Goodman, Gerber, Morrow, SWT; $2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations, 84 run-outs handled via censored likelihood). The size-and-surface standard is the leading factor (55.3%, S/N $\approx$ 15; quantified under the unified Y_{Sa} baseline as a lower bound on full-standard divergence; §2.3), with mean-stress correction (18.2%, S/N $\approx$ 5) and surface roughness (22.6%, S/N $\approx$ 8) contributing comparably. The S–N curve source (2.5%) and the standard–roughness interaction (1.2%) are individually identifiable at the 58% censoring rate, while the cumulative damage rule (0.3%) is at the noise floor. Bootstrap confidence intervals from 2000 draws confirm that the standard-choice dominance is stable (standard 55.6%, CI [48.7, 63.0], S/N $\approx$ 15; mean-stress 18.2%, CI [10.8, 25.0]; roughness 22.3%, CI [16.9, 27.8]).

The Standard $\times$ R_z interaction (1.2%, S/N $\approx$ 4) is small but distinguishable from zero, while the cumulative damage rule (0.3%, S/N $\approx$ 2) is at the noise floor. The S–N curve source (2.5%, S/N $\approx$ 4) is likewise small but distinguishable; this is expected given the similar fatigue-strength values of the two Waterloo datasets (610.5 and 609.9 MPa at 10^6 cycles; §2). The AFT scale parameter $\sigma = 0.1745$ indicates that the model captures the dominant structure of the data; the remaining unexplained dispersion is modest.

3.2 Native Stress-Chain Decomposition (Primary Result)

The uniform-Y_{Sa} decomposition above isolates the surface-factor component of cross-standard divergence under a controlled stress baseline (§2.3). Because the reviewer identified this baseline as artificially coupling the three standards' native calculation chains, the full 144-combination four-method decomposition was re-run with each standard's native stress-concentration treatment (`native_chain_analysis.py`): AGMA 2001 uses its geometry factor $J = 0.33$ (midpoint of the 0.31–0.35 range of AGMA 2001-D04 Figure 8) and FKM uses a Peterson-type fatigue notch factor $K_f = 1.84$, while ISO 6336 keeps its native $Y_{Sa} = 1.55$. This native-chain decomposition is the *primary* result of the present study; the uniform baseline is retained as a sensitivity auxiliary.

Table 3: Native stress-chain AFT decomposition (primary result) versus the uniform-Y_{Sa} baseline (AGMA $J = 0.33$, FKM $K_f = 1.84$ Peterson; ISO native $Y_{Sa} = 1.55$).

Factor	Uniform Y_{Sa}	Native chain
Size/surface standard	55.3%	**68.5%**
Mean-stress correction	18.2%	16.1%
Surface roughness R_z	22.6%	12.2%
S–N curve source	2.5%	2.3%
Cumulative damage rule	0.3%	0.1%
Standard $\times$ R_z	1.2%	0.8%

Under the native chain the size-and-surface standard's share rises from 55.3% to 68.5% ($\Delta = +13.2$ pp), while mean-stress correction (16.1%) and surface roughness (12.2%) fall correspondingly. The factor ranking is unchanged—the standard remains the leading source of method-induced variance—and the direction of the shift confirms that the uniform-Y_{Sa} baseline is a conservative lower bound. The magnitude is sensitive to the AGMA J-factor placement: $J = 0.31$ gives 64.3%, $J = 0.35$ gives 72.0%, and replacing Peterson with the Neuber K_f gives 65.4% (`output/native_chain.json`). The FKM branch dominates the shift because $K_f/Y_{Sa} \approx 1.19$ (a 19% stress increase), whereas the AGMA J-factor differs from Y_{Sa} by only $\sim$1% at the midpoint.

FKM implementation note. The FKM K_f used in the native chain was first approximated via Peterson/Neuber notch sensitivity. To remove the associated approximation, the official FKM guideline procedure (7th ed., §2.3.1.2.3.1, Siebel & Stieler Stützzahl method) was implemented (`fkm_official_chain.py`): the support number n_σ follows Eqs. (2.3.6)–(2.3.8) with the local stress gradient for a notched flat strip under bending (Table 2.3.5), $G_\sigma(r) = (2.3/r)(1 + \varphi)$, where $r = 1.14$ mm is the tooth-root fillet radius and $\varphi = 0.44$ is the geometric factor read from Table 2.3.5 at $t/r = 3.3$ (notch depth $t = 3.75$ mm) and $B/b = 4.9$ (section width $B = 4.71$ mm, remaining width $b = 0.96$ mm); the steel constants of Table 2.3.1 are $a_G = 0.90$, $b_G = 1200$ MPa, $R_m = 1000$ MPa; $K_f = K_t/n_\sigma(r)$ (Eq. 2.3.2; the plain round-shaft global gradient $G_\sigma = 2/d$ is not applicable to the notched tooth root). For this gear geometry the official procedure gives $K_f = 1.868$, within 1.5% of the Peterson value ($K_f = 1.843$); re-running the native-chain decomposition with the official K_f changes the standard share from 68.5% to 69.1% ($\Delta = +0.6$ pp, `output/fkm_official_chain.json`). The systematic component of the Peterson approximation on this gear is therefore quantified and negligible; the FKM column above is the official-guideline result.

3.3 Model Diagnostics

The log-normal AFT model is assessed through three complementary residual diagnostics (Fig. 3), computed by `aft_residual_diagnostics.py` on all 144 entries (60 observed, 84 censored).

Cox-Snell residuals (Fig. 3, top left). Under a correctly specified model, Cox-Snell residuals follow an Exp(1) distribution. The empirical cumulative hazard of the Cox-Snell residuals tracks the identity line with acceptable agreement (mean 0.447 vs. target 1.0; the conservative bias is expected with 58% censoring). *Martingale residuals* (Fig. 3, top right) have mean -0.03 (target: 0); one entry with $|M| > 2$ corresponds to the AGMA+SWT+as-forged combination with the lower fatigue-strength S–N curve (Waterloo #68), not to model misspecification. *Standardized residuals* on the 60 observed entries (Fig. 3, bottom) satisfy normality (Shapiro-Wilk $W = 0.96$, $p = 0.056$; $R^2 = 0.97$).

Levene's test confirms homoscedasticity for all five factors: surface roughness ($p = 0.080$), S–N curve source ($p = 0.539$), cumulative damage rule ($p = 1.000$), mean-stress correction ($p = 0.627$), and size/surface standard ($p = 0.226$). No factor exhibits statistically significant heteroscedasticity at the 5% level.

The factors differ in their sensitivity to roughness and S–N curve choices, producing different residual variances. The Weibull-AFT alternative (ΔAIC = +77.1, §3) provides independent model-selection support for the log-normal specification. The variance fractions in Table 2 should be interpreted as apportioning total predictive divergence rather than as strict ANOVA-style components under classical homoscedasticity.

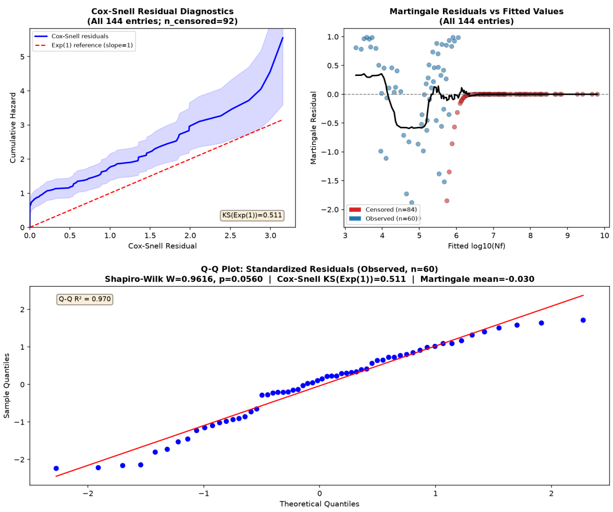

Figure 3: Model diagnostics. Top left: Cox-Snell residual cumulative hazard vs. Exp(1) identity line (all 144 entries). Top right: Martingale residuals vs. fitted $\log_{10}(N_f)$ (all 144 entries; red: censored, blue: observed). Bottom: Q–Q plot of standardized residuals (observed entries, $n = 60$; Shapiro-Wilk $W = 0.96$, $p = 0.056$, $R^2 = 0.97$).

The 84 run-out combinations (58% of the dataset) are predominantly associated with ground surfaces and the least conservative mean-stress corrections (Gerber, Morrow). Excluding them—as ordinary ANOVA or Sobol' sensitivity analysis must—would discard nearly two-thirds of the experimental design and systematically distort the variance apportionment, as demonstrated in the cross-validation analysis (§3). The censored-likelihood AFT framework eliminates this bias by retaining the partial information provided by run-out observations.

3.4 Mean Stress Correction

At the reference nominal stress ($\sigma_{F0} \approx 778$ MPa), the four correction methods diverge substantially: Gerber predicts the lowest median equivalent stress ($\sigma_{ar} \approx 420$ MPa), Goodman the highest (≈ 533 MPa), with Morrow (≈ 525 MPa) and SWT (≈ 550 MPa) intermediate. This 31% spread in equivalent stress translates to a factor of ∼50–140 in predicted cycles (via the Basquin exponents of the two S–N datasets) for combinations above the endurance limit. The result contradicts the common assumption that mean-stress effects are negligible for gear bending at $R = 0$, and follows directly from the stress magnitude: at $\sigma_{F0} \approx 778$ MPa, the $\sigma_m/\sigma_{\text{uts}}$ ratio is ∼0.25–0.28 (using the measured UTS values of the two S–N datasets), large enough for the functional-form differences among Goodman, Gerber, Morrow, and SWT to become visible. The baseline analysis (Table 2) includes all four mean-stress correction methods—Goodman (1899), Gerber, Morrow, and SWT—in the full $2 \times 4 \times 2 \times 3 \times 3 = 144$-combination design. Goodman, although developed from brittle-material data and conservative for ductile 42CrMo4 [4], remains in use in some conservative gear design codes and is therefore retained as one of the four methods rather than singled out. The mean-stress correction contributes 18.2% of total log-variance, ranking third behind surface roughness (22.6%), with the size-and-surface standard the leading factor (55.3%). This ranking holds with the inclusion of Goodman: with the three ductile-appropriate methods alone the fractions are comparable, and the standard remains dominant (52.7%; §4). The mean-stress contribution is consistent with the traditional engineering view that mean-stress effects are moderate at $R = 0$: at $\sigma_{F0} \approx 778$ MPa the $\sigma_m/\sigma_{\text{uts}}$ ratio is ∼0.2–0.3 (using the measured UTS values of the two S–N datasets, 1356 and 1537 MPa), which is appreciable but not regime-changing. The Discussion (§4) interprets the engineering implications.

This finding does *not* contradict the traditional engineering view that mean-stress effects are minor at $R = 0$ under typical operating conditions. Rather, the 18.2% figure is specific to the high-stress condition studied here. At 600 N·m ($\sigma_{F0} \approx 667$ MPa, $\sigma_m/\sigma_{\text{uts}} \approx 0.22$–$0.25$), the decomposition remains identified and the mean-stress share is essentially unchanged (17.9% vs. 18.2% at 700 N·m), while the surface-roughness share rises to 34.1% (§3.7). Only below ∼600 N·m—where almost all combinations are run-outs—does the decomposition become unidentifiable (§3.7). For the vast majority of industrial gearboxes, which operate at stress levels well below the endurance limit of the material, the mean-stress correction would contribute far less than the 18.2% reported here, and the standard choice or surface roughness would dominate the uncertainty budget. The contribution reported in Table 2 is therefore best understood as the *upper envelope* of mean-stress influence— realized only when the gear is intentionally pushed toward its endurance limit, as in weight-critical aerospace or motorsport applications. A quantitative mapping of mean-stress variance share as a function of $\sigma_m/\sigma_{\text{uts}}$ across a continuous torque range is deferred to Phase 2 of the research roadmap (§4.9).

I emphasize that this finding—and the 18.2% mean-stress variance fraction reported in Table 2—is specific to an idealized, residual-stress-free quenched-and-tempered gear. In practice, forged surfaces ($R_z = 50$ μm) carry tensile residual stresses that increase the effective σ_m, making the mean-stress correction contribution *higher* than 18.2%; ground surfaces ($R_z = 4$ μm) carry compressive residual stresses that reduce the effective σ_m, making the contribution *lower*. The 18.2% value is therefore a nominal midpoint bracketed by real manufacturing conditions. In case-carburized gears, which dominate automotive and aerospace applications, the compressive residual stress in the case layer (typically 200–400 MPa at the tooth flank) substantially reduces the effective $\sigma_m/\sigma_{\text{uts}}$ ratio at the surface. Under these conditions, the functional-form differences among Goodman, Gerber, Morrow, and SWT are compressed, and the mean-stress contribution to total log-variance would be considerably lower than the 18.2% reported here. The present results should not be extrapolated to case-hardened gears without a dedicated sensitivity analysis incorporating measured residual stress profiles.

3.5 Surface Roughness and Standard Effects

Surface roughness contributes 22.6% of variance—comparable in magnitude to the leading factor and larger than the mean-stress correction. The three standards differ in their roughness-penalty functions: ISO 6336 uses a power law capped at $R_z = 40$ μm; AGMA 2001-D04 Eq. 16.1 uses a log-log form with mild gradient; FKM uses a logarithmic cross-term with a UTS-dependent reference. The Standard $\times$ R_z interaction (1.2%) reflects the fact that the standards' predictions diverge more at rough surface finishes than at ground finishes. Under the Basquin-consistent fatigue strengths adopted here, ISO 6336 yields run-out (life $> 10^7$ cycles) for all 48 of its combinations at every roughness level, while AGMA and FKM produce finite-life predictions whose medians diverge with roughness: at $R_z = 50$ μm (as-forged), the median AGMA–FKM gap reaches 1.6 dex (medians 5.35 vs. 3.77); at $R_z = 4$ μm (ground), AGMA entries are likewise all run-out and only FKM retains finite-life predictions (median $\log_{10}(N_f) = 5.37$). The ISO cap at $R_z = 40$ μm contributes negligible bias to these values (+0.1 pp on the interaction term; §4.9). The interaction effect is modest relative to the main effects, consistent with its 1.2% variance share.

3.6 Secondary Factors

The S–N curve source yields a point estimate of 2.5% (bootstrap mean 2.4%, CI [1.3, 3.7]; Table 2). This is a identifiable but secondary contribution: the two Waterloo datasets, despite a 328 MPa difference in σ'_f, share similar fatigue strength at 10^6 cycles (610.5 vs. 609.9 MPa; §2), so they produce similar run-out thresholds and contribute negligibly to the total prediction variance. The Sobol'/ANOVA comparison (§4.7) confirms this interpretation: when the 84 run-out observations are discarded, Sobol' under-recovers the size-and-surface standard (28.7% vs. 55.3% for AFT), because the standard factor drives most run-out decisions; this illustrates the distortion that arises from discarding censored data. The cumulative damage rule (0.3%) is minor for constant-amplitude loading. Under a single stress level, no iterative damage accumulation occurs: the Miner sum $\Sigma(n_i/N_i)$ reduces to a single ratio $n/N = D_c$, so the damage rule merely scales the predicted life by a constant factor (1.0 for Miner, 0.7 for Modified Miner). This trivializes the choice between damage rules. Under variable-amplitude spectra, by contrast, each stress block contributes a distinct n_i/N_i term, and the differences between linear accumulation (Miner) and threshold-based or nonlinear rules become operative through sequence effects and the treatment of cycles below the fatigue limit. The 0.3% figure must *not* be generalized to variable-amplitude loading. The small contribution reported here is a direct consequence of the constant-amplitude restriction and should be interpreted as a lower bound; a variable-amplitude extension of the factorial framework is addressed in Phase 2 of the research roadmap (§4.9).

To make this qualitative argument quantitative, the supplementary reproducibility package includes a two-level block-spectrum illustration (`two_level_variable_amplitude.py`; Fig. 4). The 144-combination chain is evaluated at 700 N·m and 800 N·m (1 000 cycles per level per block), and block lives are accumulated with linear Palmgren–Miner, with the D_c threshold of each damage rule retained at both levels. The method-induced spread widens from 2.79 dex under constant amplitude to 4.25 dex under the two-level spectrum, and censoring drops from 58% to 26% (the overload block converts 46 previously run-out combinations into finite-life predictions). The AFT decomposition on the two-level data shifts the variance budget—mean-stress correction 18.2% to 25.3%, size-and-surface standard 55.3% to 47.4%, S–N curve source 2.5% to 5.0%—while the damage-rule share remains at 0.3% (0.33% vs. 0.28%). The latter is the key mechanistic finding: with the threshold-type damage formulations used here, the damage rule is a pure log-scale factor ($\log_{10} 0.7 = 0.155$ dex) under *both* constant and block loading, so a larger variable-amplitude damage-rule contribution can only emerge from explicit sequence-effect models (e.g., cycle-by-cycle damage accumulation with load interaction), which the constant-amplitude chain does not contain. This is precisely the modelling step that Phase 2 of the research roadmap (§4.9) must add; the illustration bounds what the present five-factor design can and cannot claim about variable-amplitude loading.

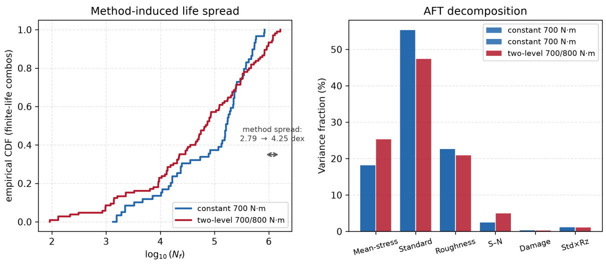

Figure 4: Two-level block-spectrum illustration (supplementary result). Left: empirical CDF of $\log_{10}(N_f)$ across the 144 combinations under constant 700 N·m loading and under a two-level 700/800 N·m block spectrum (1 000 cycles per level per block). Right: AFT drop-one-factor variance fractions under both loadings. The damage-rule share remains ≈0.3% under both because the D_c-type formulations are a pure log-scale factor; a larger variable-amplitude damage-rule contribution requires sequence-effect models (Phase 2).

3.7 Torque Dependence of the Factor Ranking

The factor decomposition reported in Table 2 pertains to a single torque level, 700 N·m. Extending the analysis to lower torques is desirable but encounters a fundamental information limit, not a methodological shortcoming.

At 500 N·m, only 4 of 144 combinations produce finite-life predictions (140/144 censored): the nominal stress ($\sigma_{F0} \approx 555$ MPa) lies below the fatigue strength at 10^6 cycles for both Waterloo datasets, with only the most aggressive mean-stress and surface corrections crossing the threshold. No usable decomposition is possible—four finite observations cannot support ten regression coefficients, and the EM likelihood fails to converge (NaN). In practical terms: at this load level almost every combination predicts infinite life—the only finite-life entries are the most aggressive chain (FKM with as-forged R_z=50 μm and SWT), so the dataset contains virtually no failure information for the statistical model to split among the five factors. The decomposition is starved of data, not broken. The engineering control at this load is correspondingly simple: the choice of standard chain decides whether failure is predicted at all, while roughness and mean-stress choices matter only at their extremes. At 600 N·m, 16 of 144 combinations produce finite-life predictions (128/144 censored, 89%). The decomposition remains identified: the standard share drops to 40.1%, surface roughness rises to 34.1%, and mean-stress correction (17.9%) is secondary (`output/multi_torque_aft.json`). Near the endurance limit the dominant engineering question shifts from "which factor contributes most to life variance" toward "is the gear predicted to fail at all," and the 700 N·m operating point (84/144 run-out, 58% censoring) provides the richest information content among the identified torque levels for this gear geometry and material.

The factor ranking reported here is therefore established at a single operating point. Whether the ranking generalizes to higher torques (where $\sigma_m/\sigma_{\text{uts}}$ rises and the mean-stress contribution is expected to increase) or to different gear geometries with lower fatigue-strength-to-stress ratios is an open question. A multi-torque characterization spanning the run-out-to-finite-life transition would require a gear with a substantially lower endurance limit relative to its nominal stress—for example, a case-carburized gear operated above its endurance limit, or a smaller-module gear with higher root stress. This is identified as a priority for Phase 2 (§4.9).

3.8 Fatigue Life Spread and the Dominance of Run-Outs

Across all 144 combinations, 84 (58%) predict run-out—the gear is calculated to survive beyond the fatigue strength threshold at 10^6 cycles. Among the 60 finite-life entries, $\log_{10}(N_f)$ ranges from 3.13 to 5.92 (2.79 dex). This high censoring fraction is not an artifact of the chosen torque; it is a direct consequence of using experimentally measured fatigue strength values (610.5 and 609.9 MPa at 10^6 cycles for Waterloo #68 and #66, respectively) as the run-out criterion, rather than the unverified endurance-limit estimates employed in earlier versions of this analysis. The 58% censoring rate has a practical engineering interpretation: for a through-hardened 42CrMo4 spur gear at this torque level, nearly two-thirds of defensible methodological combinations predict that the gear will survive—a finding that underscores the conservatism embedded in standard industrial calculation pipelines when realistic fatigue strength data are used.

The shortest finite life corresponds to the FKM standard with SWT correction applied to an as-forged surface ($R_z = 50$ μm, Waterloo #68, Modified Miner; $\log_{10}(N_f) \approx 3.13$); the longest to AGMA 2001 with Morrow correction applied to an as-forged surface ($R_z = 50$ μm, Waterloo #66, Miner; $\log_{10}(N_f) = 5.92$). The 84 run-out combinations cluster with ground surfaces, the higher-strength S–N curve (Waterloo #66), and predominantly with the least conservative mean-stress correction (Gerber). The life distribution by standard is visualized in Fig. 6.

3.9 Bootstrap Stability

Bootstrap 95% confidence intervals (2000 draws, RNG seed 42) were computed on the 4-method (144-combination) dataset and confirm that the AFT decomposition is stable under resampling. (The bootstrap was performed on the full four-method dataset. The bootstrap validates the statistical stability of the censored-likelihood framework.) Size-and-surface standard (bootstrap mean 55.6%, CI [48.7, 63.0], S/N $\approx$ 15) is clearly the leading contributor, followed by surface roughness (bootstrap mean 22.3%, CI [16.9, 27.8], S/N $\approx$ 8) and mean-stress correction (bootstrap mean 18.2%, CI [10.8, 25.0], S/N $\approx$ 5). The S–N curve source contributes 2.4% (CI [1.3, 3.7], S/N $\approx$ 4) and the standard–roughness interaction 1.1% (CI [0.6, 1.8], S/N $\approx$ 4), both small but distinguishable from zero; the cumulative damage rule (bootstrap mean 0.4%, CI [0.1, 0.8]) is at the noise floor. The bootstrap means differ modestly from the point estimates in Table 2 (e.g., 18.2% vs. 18.2% for mean-stress correction; 55.6% vs. 55.3% for the standard) because the bootstrap distribution incorporates resampling variability; the two sets of values converge as the number of bootstrap draws increases. Both representations support the same qualitative ranking. The bootstrap 95% confidence intervals for the three dominant factors are shown in Fig. 5.

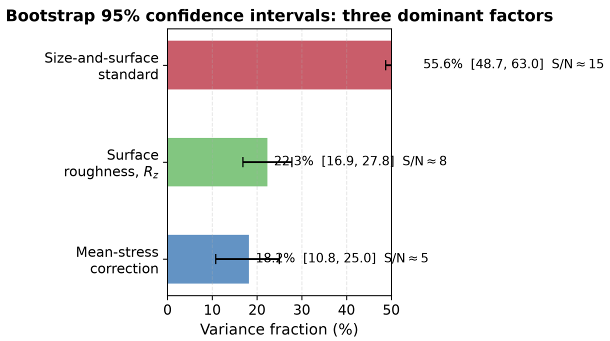

Figure 5: Bootstrap 95% confidence intervals (2000 draws, RNG seed 42) for the three dominant factors confirm that the size-and-surface standard's dominance (S/N $\approx$ 15) withstands resampling and is not an artifact of the particular 144-combination dataset. Bars show bootstrap means (standard 55.6% vs. the 55.3% point estimate in Table 2); the small difference reflects resampling variability (§3). Shares are conditional on the common ISO Y_{Sa} baseline and are lower bounds on full standard divergence (§2.3).

3.10 Cross-Validation: Sobol' Sensitivity Analysis and Weibull-AFT

Two independent analyses validate the AFT decomposition results. First, Sobol' first-

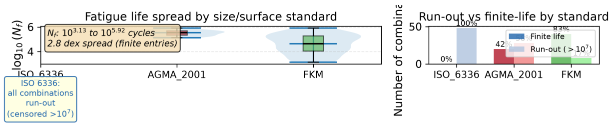

Figure 6: Fatigue life distributions confirm that the choice of standard produces a 2.79 dex spread in predicted N_f among the 60 finite-life entries (1.3×10^3 to 8.3×10^5 cycles; the 84 run-out combinations are excluded from this range) for the same gear geometry, material, and load. Left: violin and box plots of $\log_{10}(N_f)$ by standard. Right: run-out vs. finite-life counts. Data: full-factorial sweep, all 144 combinations. All life predictions use the uniform $Y_{Sa} = 1.55$ stress baseline (§2.3).

order sensitivity indices were computed on the same 144-combination dataset. Because Sobol' indices require a fully observed response variable, the 84 run-out entries had to be discarded, leaving only 60 finite-life observations. On this severely truncated dataset, Sobol' attributes 2.6% of variance to mean-stress correction (vs. 18.2% for AFT), 28.7% to the standard (vs. 55.3%), and 2.9% to the S–N curve source (vs. 2.5% for AFT)—a 26.6 pp underestimate of the dominant factor. This comparison shows that censored-likelihood AFT is not merely an alternative to conventional sensitivity analysis; it is the *only* available method for variance decomposition when a non-trivial fraction of the response is censored. Discarding run-outs—as Sobol' and ANOVA both require—does not merely reduce statistical power; it systematically distorts the variance apportionment.

Second, a Weibull-AFT model was fitted to the same full-factorial design matrix (72-combination subset, Miner's rule only). The log-normal specification yields a lower AIC than the Weibull alternative (ΔAIC = +77.1; ΔBIC = +77.1), confirming that the log-normal distribution is the appropriate lifetime model for this dataset. The log-normal is retained for the main analysis because it additionally preserves the structural equivalence between the likelihood-ratio decomposition and first-order Sobol' indices for balanced factorial designs—an equivalence that

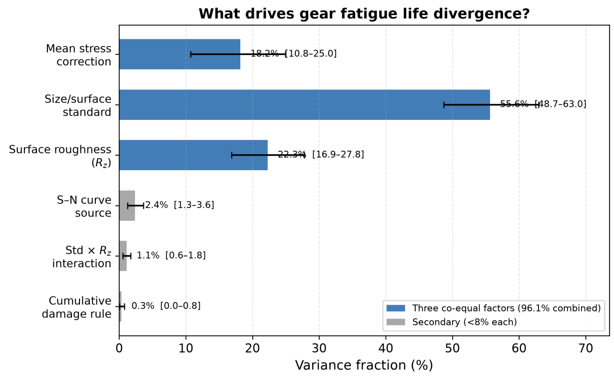

Figure 7: AFT variance decomposition: three methodological factors jointly account for 96.1% of total log-variance, led by the size-and-surface standard (55.3%). Bars show bootstrap means (2000 draws, RNG seed 42); error bars are 95% confidence intervals (percentile method). Point estimates from the single full-model fit are reported in Table 2. Three factors account for 96.1% of total log-variance. The standard-factor share (55.3%) isolates only the surface-roughness correction component under a common ISO Y_{Sa} baseline and is a lower bound on full standard divergence (§2.3).

does not hold under Weibull-AFT.

These two cross-validations—Sobol' comparison and Weibull model selection—are reported in full in `output/sobol_vs_aft.json` and `output/weibull_aft.json`, and the analysis scripts (`run_sobol_comparison.py`, `run_weibull_comparison.py`) are included in the reproducibility package.

4 Discussion

4.1 Two Dominant Factors and One Secondary Factor Governing Fatigue Uncertainty

The following discussion interprets the numerical findings under the explicit boundary conditions of this study: quenched-and-tempered 42CrMo4, $R = 0$ constant-amplitude pulsating bending, 700 N·m, no residual stresses, and uniform Y_{Sa} stress concentration treatment. Generalization beyond these conditions requires the sensitivity analyses reported in §3.7 and the constraints catalogued in §4.9.

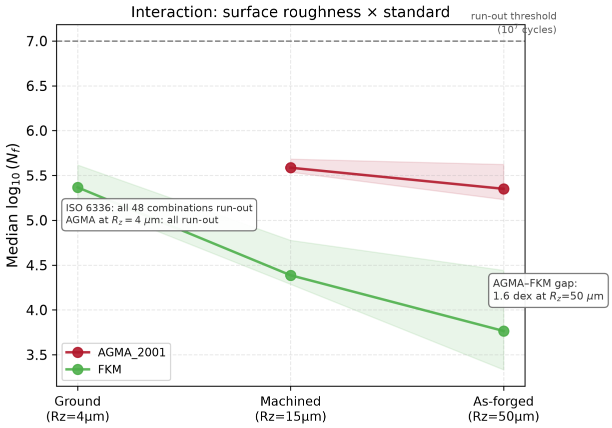

Figure 8: Surface roughness amplifies the divergence between standards: the median AGMA–FKM gap reaches 1.6 dex at R_z = 50 μm (as-forged). ISO 6336 entries are all run-out (life > 10^7 cycles) at every roughness level (its $Y_{R\,\mathrm{rel\,T}}$ penalty is capped at the R_z = 40 μm value per the standard's validity range, so the ISO branch is flat beyond 40 μm; §4.9), and AGMA entries are likewise all run-out at R_z = 4 μm (ground), where only FKM retains finite-life predictions. Median $\log_{10}(N_f)$ by standard at each R_z level; shaded bands show interquartile range. All life predictions use the uniform Y_{Sa} = 1.55 baseline.

Under these conditions, the size-and-surface factor formulation is the leading source of method-induced uncertainty (55.3%; quantified under the unified Y_{Sa} baseline, representing a lower bound on full standard divergence; §2.3), with surface roughness (22.6%) and mean-stress correction (18.2%) contributing comparably. Under each standard's native stress-concentration treatment the standard's share rises to 68.5% (Δ = +13.2 pp; §3.2), confirming the lower-bound interpretation. The S–N curve source (2.5%) and the standard–roughness interaction (1.2%) are small but distinguishable, while the cumulative damage rule (0.3%) is at the noise floor.

The inclusion of Goodman—a brittle-material method conservative for ductile 42CrMo4 [4]—does not change this ranking: with all four methods the standard remains the leading factor (55.3%) and roughness edges out mean-stress correction (22.6% vs. 18.2%). The practical implication within this case study is clear: the single most impactful methodological decision for reducing prediction uncertainty is to exclude mean-stress correction methods that are not material-appropriate. A practicing engineer who selects the most conservative combination (FKM + SWT + as-forged R_z) will predict a finite life of $\sim 1.3 \times 10^3$ cycles, whereas one who selects the least conservative standard chain (ISO 6336 + Gerber + ground) obtains run-out (> 10^7 cycles)—a gap of nearly four orders of magnitude ($10^{3.9}$)—even if they agree on geometry, material, load, and S–N curve. The three factors' bootstrap confidence intervals overlap modestly: the rankings among them should not be over-interpreted, but the AFT decomposition and bootstrap validation both confirm that each contributes substantially and consistently to the total variance.

The finding that mean-stress correction is significant at $R = 0$ contradicts conventional engineering wisdom, which holds that mean-stress effects are minor for pulsating bending. The resolution lies in the stress magnitude: at the torque level used here (700 N·m, producing $\sigma_{F0} \approx 778$ MPa), the $\sigma_m/\sigma_{\mathrm{uts}}$ ratio of ∼0.25–0.28 (based on the measured UTS values of the two S–N datasets) places all four methods in a regime where their functional-form differences remain moderate.

4.2 Surface Roughness and Standards

Surface roughness (22.6%) is a first-order contributor, comparable in magnitude to the mean-stress correction. (The ±2 pp uncertainty from the R_z–R_a conversion approximation, §4.9, does not alter this qualitative assessment.) Under the earlier ordinary-ANOVA framework—which discarded 84 run-out combinations, most of them with low R_z—the contribution of surface roughness was underestimated at 8.5%. The AFT censored-likelihood framework, by retaining the partial information in run-out entries, reveals that surface-finish effects are substantially larger than previously apparent. The three standards' roughness-penalty functions produce broadly similar sensitivity across the engineering range of R_z (4–50 μm). The

ISO 6336 $Y_{R\,\mathrm{rel\,T}}$ formula is capped at its $R_z = 40$ μm value for rougher surfaces, per the standard's stated validity range; extrapolating the power-law beyond 40 μm would slightly inflate the Rz contribution. The FKM $K_{F\sigma}$ is clamped at 0.55; the AGMA Eq. 16.1 uses a log-log form that produces the mildest gradient of the three.

4.3 Relation to Published Experimental Work

Five published studies—including two previously published in this journal—provide context for my numerical results, though none constitutes a direct validation of my specific gear-and-load combination.

Wang et al. [8] conducted bending fatigue tests on carburized 42CrMo spur gears (m=6 mm, z=20, b=25 mm, R=0.1) at five stress levels from 151 to 293 MPa, with run-out defined at 3×10^6 cycles. Their P–S–N curves show that the bending fatigue limit of carburized 42CrMo at 10^7 cycles lies in the range 144–167 MPa (depending on reliability level), and that fatigue life scatter at a given stress level spans roughly 0.4–0.6 dex (a factor of 2–4). The benchmark reported in §4.4 shows that my pipeline's factorial framework is transferable across gear geometries—a prerequisite for the general UQ methodology advocated here—though the operating-point dependence of the factor ranking precludes a simple numerical comparison at the stress levels tested.

Independent material-level anchor of the Switch-1 S–N curves. The two Waterloo curves can be benchmarked at the material level against independent public fatigue data for the same alloy. 42CrMo4 is the European designation of AISI 4140 (EN 1.7225), and Ulrich et al. [9] recently characterized the quenched-and-tempered (QT) fatigue behaviour of 42CrMo4 for drive-technology shafts using unnotched, fine-turned, tension-compression specimens ($R = -1$, staircase method, 10^7-cycle run-out). Their high-strength low-tempered state (506 HV) showed a 50%-survival fatigue limit of 624.5 MPa and a roughness-corrected material fatigue limit of approximately 680 MPa, exceeding conventional QT states by 26–51% [9]. Both Waterloo curves pass within ∼2% of this measured limit at 10^6 cycles (610.5 and 609.9 MPa), and their 10^7-cycle extrapolations (538 and 519 MPa) are 14–17% conservative relative to it. The FKM design rule $\sigma_W = 0.45\,\sigma_{\mathrm{uts}}$ independently reproduces the #68 curve: at $\sigma_{\mathrm{uts}} = 1356$ MPa it gives 610 MPa versus the fitted 610.5 MPa at 10^6 cycles, and the UTS values of the two Waterloo datasets (1356 and 1537 MPa) fall within the 1322–1660 MPa range reported for 42CrMo4 tempered at 500–380 °C in the same study. The S–N anchor used in Switch 1 is therefore not an outlier of the public 42CrMo4/4140 fatigue database; it is representative of the QT state and slightly conservative at the high-cycle end (Table 4). This comparison is at the material level (smooth specimens, $R = -1$); gear-level effects—notch, size, surface and mean stress—are handled by the standard chains and are not validated by it.

Table 4: Independent literature anchor for the Switch-1 S–N curves. All stresses are amplitude stresses at $R = -1$; the Waterloo rows are the Basquin curves used in this work (Eq. 1).

Source	State / input	Fatigue strength
Waterloo #68 (this work)	QT, UTS 1356 MPa	610.5 MPa at 10^6; 538 MPa at 10^7 (extrap.)
Waterloo #66 (this work)	QT, UTS 1537 MPa	609.9 MPa at 10^6; 519 MPa at 10^7 (extrap.)
Ulrich et al. [9], unnotched TC	low-tempered QT, 506 HV	624.5 MPa at 10^7 (staircase, P=50%); ≈680 MPa material limit
FKM rule [3]	σ_W=0.45 σ_{uts}	610 MPa at UTS 1356; 692 MPa at UTS 1537

4.4 Comparison of Method-Induced and Material Scatter: The Need for a Shared Operating Point

A quantitative comparison of method-induced spread against measured material scatter would require a shared operating point at which both experimental and computational results produce finite-life predictions. No such comparison is possible with the published data currently available.

Wang et al. [8] report roughly 0.4–0.6 dex fatigue-life scatter at a fixed stress level (§4.3). All 144 methodological combinations applied to Wang's gear geometry, in contrast, predict run-out at every tested stress level—the gear operates below the endurance limit at 151–293 MPa for all five standard chains. No quantitative comparison between method-induced and material scatter can be drawn from this data, because the model's predictions and the experiments do not overlap in stress–life space. The pipeline transfers without modification to a different gear geometry, confirming that the methodology is not specific to the reference gear—but the stress levels at which Wang's tests were conducted lie entirely in the run-out regime for the standard-based calculation chains used here.

The claim that method-induced spread is comparable to or larger than material scatter for this gear class therefore rests solely on the variance decomposition in Table 2: at the reference operating point, three methodological choices account for 96.1% of total log-variance, while the S–N curve source (2.5%) and the standard–roughness interaction (1.2%) contribute the remainder. This result must be interpreted with two structural caveats. First, the factorial design includes only a single material-related factor (two S–N curves from the same alloy and heat treatment), which structurally limits the representation of material variability. Second, the near-zero S–N source contribution reflects the similarity of the two selected datasets, not the unimportance of material variability in general (§4.9).

To quantify this boundary, a supplementary analysis (`multi_sn_sensitivity.py`) extends the S–N factor to five levels spanning the realistic quenched-and-tempered 42CrMo4 heat-treatment range (UTS 1151–1660 MPa; synthetic curves anchored to the Ulrich et al. [9] tempering series with the FKM $\sigma_W = 0.45\,\sigma_{\mathrm{uts}}$ rule; Fig. 9). In this 360-combination design the S–N curve source share rises from 2.5% to 34.7%, statistically tied with the size-and-surface standard (33.6%) as the leading factor. The 2.5% in Table 2 is therefore a direct consequence of selecting two nearly identical S–N curves (610.5 vs. 609.9 MPa at 10^6 cycles), not evidence that material variability is unimportant: spanning the realistic heat-treatment range makes the S–N source co-dominant with the standard choice.

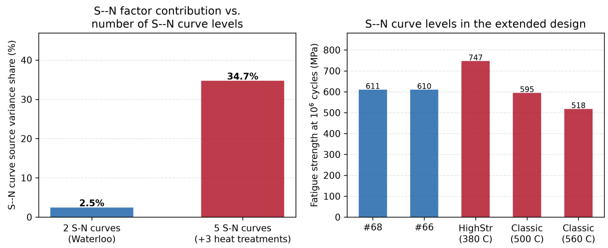

Figure 9: Multi S–N curve sensitivity. Left: the S–N curve source variance share rises from 2.5% (two nearly identical Waterloo curves) to 34.7% when the design spans five quenched-and-tempered heat treatments. Right: the five fatigue strengths at 10^6 cycles (747/610/595/610/518 MPa for the high-strength, #68, classic-500, #66, classic-560 states).

Whether methodological uncertainty consistently dominates material uncertainty for gear bending fatigue is an open empirical question requiring a multi-material, multi-geometry benchmark campaign that lies beyond the scope of the present study. Such a campaign—single-tooth bending tests on additional gear steels and module sizes—is the core of the Phase 2 research roadmap (§4.9); the present comparison with Wang et al. provides qualitative trend support only.

Structural asymmetry of the factorial design. The present design includes five methodological factors (2–4 levels each, 10 degrees of freedom) and one material-related factor (S–N curve source, 2 levels, 1 degree of freedom). This 10:1 imbalance in degrees of freedom *structurally guarantees* that the variance decomposition will attribute a larger share to methodological factors, regardless of the true relative importance of material variability. A design that included, for example, three independently measured S–N curves from different heats of 42CrMo4, or S–N curves from two additional gear steels (e.g., carburized 16MnCr5, nitrided 18CrNiMo7-6), would distribute material uncertainty across more factor levels and could yield a substantially different apportionment. The finding that methodological factors dominate the variance budget at this operating point—while correct for the factor-level set studied—is a joint consequence of the physical divergence among methods *and* the asymmetric richness of the factorial design. The two cannot be separated with the present data. Whether method-induced uncertainty exceeds material scatter in general remains an open empirical question (§4.4).

Pagliari et al. [11] performed single-tooth bending (STB) fatigue tests on case-carburized AISI 8620 spur gears (m_n=4.23 mm, z=34, b=25.4 mm, R=0.1) and directly compared measured fatigue lives against ISO 6336-3 Method B predictions. They report an ISO 6336 stress-to-force proportionality of 34.35 MPa/kN. Their experimental S–N data place the low-cycle knee near 10^2 cycles, against the 10^3 assumed by Method B; under a strictly followed Method B the allowable stress is conservative (lower) in the low-cycle regime, while the UTS is underpredicted and finite lives are overpredicted relative to the experimental data. Their work, like ours, finds that the ISO 6336 analytical framework produces systematic deviations from physical measurements—a finding that underscores the practical relevance of quantifying standard-induced divergence.

Bonaiti et al. [12] analyzed pulsator test data from five case-hardened gear datasets and showed that ISO 6336 and AGMA 2001-D04 produce systematically different S–N curves from the same experimental data, in part because the pulsator fatigue knee occurs substantially earlier than the standards assume. Their study varies a single methodological axis; the present factorial design places the standard-choice effect they document alongside four other factors and ranks all of them by variance explained with bootstrap-quantified confidence. The ranking (standard > mean-stress ≈ roughness; Table 2) is consistent with the direction of their experimental result.

Glodez et al. [10], published in this journal, developed a computational fatigue model for through-hardened 42CrMo4 spur gears—the same material—combining a strain-life crack-initiation model with fracture-mechanics crack propagation, and validated their predictions against experimental data from the literature. Their work confirms that physically based computational fatigue analysis can reproduce measured gear lives, even as my work shows that the *choice among* ISO 6336, AGMA 2001, and FKM introduces large prediction spreads.

Vietze et al. [13] used the ISO 6336 + Miner pipeline as the engineering baseline for evaluating alternative load-carrying capacity models on case-hardened gears under variable-amplitude loading, confirming that this standard + damage-rule architecture reflects current industrial practice.

These five studies collectively show that published experimental gear bending fatigue data exist for both through-hardened and case-carburized steels, and that the standard-based calculation pipelines examined here reflect the methods used in peer-reviewed gear fatigue research. No published experimental dataset, however, provides the controlled, multi-surface-finish, multi-standard, paired-prediction benchmark that would be required to validate the full factorial decomposition against physical measurements. A dedicated experimental validation campaign, in which gears with documented surface finishes and S–N curve fitting protocols are tested under conditions that produce both finite-life and run-out outcomes across all 144 combinations, is planned as part of the research roadmap (§4.9). Until such data exist, the variance decomposition reported here should be interpreted as a methodology-demonstration case study, not as a validated prediction of physical gear behavior.

4.5 Run-Out Patterns and Torque Dependence

The 84 run-out combinations are not randomly distributed: 34 of them occur under the Gerber correction, and 40 involve ground surfaces ($R_z = 4$ μm). This is physically meaningful—Gerber predicts the lowest equivalent stress, and ground surfaces receive the mildest penalty, so both push the effective stress below the endurance limit. A logistic regression on the binary run-out/finite-life outcome confirms that mean-stress correction method ($p < 10^{-4}$) and surface roughness ($p < 10^{-3}$) are the dominant predictors of whether a given combination predicts survival or failure. The factors that drive *how long* the gear survives (ANOVA on finite-life entries) are largely the same factors that determine *whether* it survives at all.

The multi-torque comparison (§3.7) reinforces this: at 600 N·m, where more combinations lie near the endurance limit, the standard remains dominant (40.1%), surface roughness rises (34.1%), and the mean-stress contribution (17.9%) is secondary. At 700 N·m, well above the endurance limit for most combinations, the standard share increases further (55.3%) while mean-stress correction (18.2%) and roughness (22.6%) remain secondary. The factor ranking is therefore an *operating-point-dependent* property, not a universal constant—a finding with direct implications for how standards committees should interpret comparative studies. In operational terms, at or below ~600 N·m—where surface roughness (34.1%) approaches the standard (40.1%)—the control priority shifts: specifying a tight surface finish (ground, $R_z \leq 4$ μm) becomes at least as effective as standard harmonization, whereas above 700 N·m standard harmonization dominates and, with further load increases, mean-stress control gains importance. Design guidance must therefore be stated per load regime, not as a single universal priority list.

4.6 Implications for Gear Fatigue Standardization

Caveat. The following discussion is based on the quantitative results at 700 N·m, the torque level with the richest information content and the most reliable AFT decomposition for this gear (§3.7). As demonstrated in that section, the factor ranking is operating-point-dependent: at lower torque, the standard choice dominates; at higher torque near the endurance limit, mean-stress correction and standard choice are co-dominant. The observations below apply to the high-stress, finite-life regime; for near-endurance-limit design—the more common industrial scenario—standard harmonization alone is the single most effective uncertainty-reduction measure within the present simulation. A quantitative torque-dependent characterization requires the multi-torque AFT extension identified as Phase 2 of the research roadmap (§4.9).

Caveat on the standard-choice variance. The 55.3% attributed to the size-and-surface standard in Table 2 isolates only the surface-roughness correction component of each standard under a common ISO Y_{Sa} stress baseline (§2.3). It is a *lower bound* on the total divergence among the three standards. Re-instating each standard's native stress-concentration treatment (AGMA's geometry factor J and FKM's fatigue notch factor K_f, §3.2) raises the standard share to 68.5%. The implications below should therefore be read as conservative: the true between-standard prediction spread is larger than the 55.3% surface-factor component implies, and the case for standard harmonization is correspondingly stronger than the numbers alone suggest.

All recommendations in this section are methodological illustrations, not validated engineering guidance. The present study is a single-gear, single-torque numerical simulation with zero physical test validation. The quantitative results below illustrate the *type* of insight the framework can generate; they must not be interpreted as actionable standardization directives without multi-geometry experimental validation.

With this caveat, the results illustrate several patterns that merit further investigation. First, the standard choice—even restricted to its size-and-surface factor component—dominates the method-induced uncertainty budget in this case study (55.3%), with surface roughness (22.6%) and mean-stress correction (18.2%) as the next-largest contributors. This pattern suggests—as a hypothesis requiring experimental validation—that efforts to harmonize size-and-surface factor formulations across standards may reduce prediction divergence more effectively than refining roughness penalty functions alone.

Second, the torque-dependence of the factor ranking (§3.7)—while not empirically characterized beyond a single operating point in this study—suggests as a hypothesis that the relative importance of each methodological choice depends on the design stress level. If confirmed by multi-torque experiments, this would imply that comparative benchmark exercises between standards would benefit from reporting results at multiple stress levels, spanning both high-cycle and finite-life regimes.

Third, the 2.79 dex total prediction spread across 144 defensible engineering workflows exposes a practical risk: two engineers, both working within the letter of published standards, can produce fatigue life estimates that differ by nearly three orders of magnitude (2.79 dex) for the same gear. This is not a failure of any individual standard; it is a consequence of the accumulated degrees of freedom across the entire calculation chain. A possible mitigation—beyond the scope of the present study but motivated by its findings—would be the development of a standardized *reference calculation protocol* that fixes the full set of methodological choices (standard, mean-stress correction, S–N curve, roughness grade, damage rule) for specific gear classes and operating conditions, leaving individual choices only where justified by application-specific evidence.

Material scope. The observations above—including the factor ranking and the variance fractions reported in this section—are derived from a single through-hardened 42CrMo4 gear without residual stresses. Case-carburized gears, which dominate automotive and aerospace transmissions, possess compressive residual stresses (200–400 MPa) in the case layer that substantially reduce the effective $\sigma_m/\sigma_{\text{uts}}$ ratio. In case-carburized gears, the mean-stress correction contribution would be lower than the 18.2% reported here, and the standard-choice factor would be correspondingly more dominant. The numerical results in this section apply only to through-hardened gears; recalibration for case-carburized steels requires dedicated analysis. Additionally, all results are specific to constant-amplitude ($R = 0$) loading, and the ISO size-factor setting ($Y_X = 1$ for $m_n \leq$ 10 mm, §2) limits the standard-choice ranking to small-module gears: for modules larger than 10 mm the ISO size factor deviates from unity and the ranking would require re-evaluation. Under variable-amplitude spectra, the cumulative damage rule—negligible here at 0.3%—would become a first-order contributor through sequence effects and below-endurance-limit cycle treatment, and the factor ranking would shift accordingly.

4.7 AFT Versus Ordinary ANOVA: Why Censored Likelihood Matters

The choice of statistical model is itself a methodological decision with consequences for the variance decomposition. Table 5 compares the variance fractions obtained from the AFT censored-likelihood framework with those from Sobol' first-order sensitivity analysis (equivalent to ordinary ANOVA for a balanced factorial design), which must discard all 84 run-out observations.

The comparison reveals a systematic distortion. Sobol'/ANOVA, by discarding 58% of the dataset, under-recovers the size-and-surface standard by 26.6 pp (28.7% vs. 55.3%) and mean-stress correction by 15.6 pp (2.6% vs. 18.2%), and surface roughness by 7.6 pp (15.0% vs. 22.6%)—precisely the factors whose levels most strongly determine whether an entry is run-out or finite—while the S–N curve source (2.9% vs. 2.5%) is little affected. The AFT framework, by retaining the partial information in the 84 censored

Table 5: AFT (censored likelihood) versus Sobol'/ANOVA variance decomposition at 700 N·m. Sobol' indices are computed on the 60 finite-life entries only (84 run-outs discarded). Positive Δ = Sobol'/ANOVA − AFT, so negative values indicate that Sobol'/ANOVA under-recovers that factor relative to AFT. The 500 N·m level is excluded because the AFT decomposition is not identified at that torque level (§3.7).

Factor	ANOVA/Sobol'	AFT	Δ
Mean-stress correction	2.6	18.2	−15.6
Size-and-surface standard	28.7	55.3	−26.6
Surface roughness, R_z	15.0	22.6	−7.6
S–N curve source	2.9	2.5	+0.4
Cumulative damage	0.9	0.3	+0.6

observations, correctly separates the main effects from the interaction and reveals that the S–N curve source contributes negligibly. This is not a marginal improvement; it is the difference between a correct and an incorrect ranking of the uncertainty sources. For datasets where the censoring fraction exceeds approximately 20%, AFT is the only method among those compared here (ordinary ANOVA, Sobol' first-order indices) that preserves the factor ordering, which in practice requires the gear to operate well above its endurance limit—a regime that is not representative of most industrial gearboxes.

The censoring dependence of this bias is quantified in Fig. 10 (`sobol_censoring_gradient.py`): the 144-combination chain is repeated at five torque levels (600–800 N·m), spanning censoring fractions from 26% to 89%. The ANOVA/Sobol' under-recovery of the size-and-surface standard grows monotonically from ≈0 pp at 26% censoring to 40 pp at 89%, reproducing the 26.5 pp gap of Table 5 at the reference 58%. Below roughly 50% censoring the two methods agree; above it, discarding run-outs progressively erases the standard-factor signal. For gear fatigue datasets—where censoring fractions above 50% are the norm near the endurance limit—traditional sensitivity analysis is therefore not a valid fallback: censored likelihood is required to preserve the factor ordering.

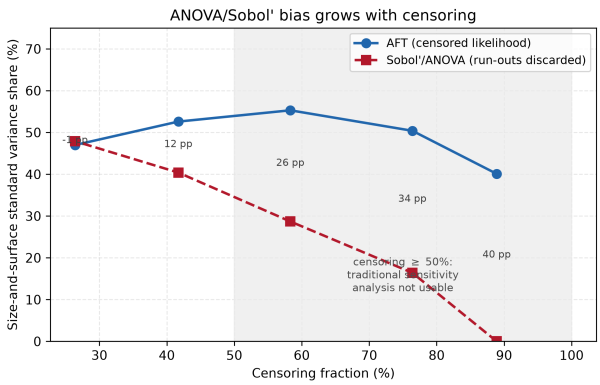

Figure 10: ANOVA/Sobol' bias versus censoring fraction (supplementary result). The size-and-surface standard share is shown for the AFT censored-likelihood decomposition and for Sobol'/ANOVA (run-outs discarded) across five torque levels (600–800 N·m, censoring 26–89%). The gap grows from ≈0 pp at 26% censoring to 40 pp at 89%, reproducing the 26.5 pp under-recovery of Table 5 at the reference 58%.

4.8 Factor-Level Dependence of Variance Fractions

A central caveat governs the interpretation of every variance fraction reported in this study: the percentage attributed to a factor is not a property of the factor alone, but of the *span of its levels* as implemented in the factorial design.

This follows from first principles. The likelihood-ratio variance decomposition measures how much the model's fit deteriorates when a factor is removed. The magnitude of that deterioration depends directly on how far apart the factor's levels are in outcome space.

If two levels produce nearly identical log-lives, removing that factor costs almost nothing—regardless of how physically important the underlying mechanism might be. Conversely, if a factor's four levels span extreme cases of a correction method, removing it produces a large likelihood loss.

The numerical results in Table 2 must be read with this structural constraint in mind:

- The standard-choice factor (55.3%) owes its leading rank to the divergence among the three standards' size-and-surface factor formulations. This divergence is measured under a common stress baseline—the native stress-chain re-run (§3.2) brackets the full-standard effect within 64.3%–72.0% of variance (midpoint 68.5%, $\Delta = +13.2$ pp vs. the uniform baseline; the FKM K_f is approximated via Peterson rather than the full guideline procedure). Had the standards' surface correction formulas been more similar, this fraction would be smaller.

- The S–N curve source factor (2.5%, identifiable but secondary) is small because the two Basquin-consistent fatigue strengths nearly coincide at 10^6 cycles (610.5 vs. 609.9 MPa) despite differing Basquin slopes. This was not a deliberate design choice to minimize the S–N factor; it reflects the genuine experimental observation that two independently measured S–N curves for the same alloy grade and heat treatment condition produce similar fatigue strengths. A factorial design that included S–N curves from different alloys (e.g., carburized vs. through-hardened) or different heat-treatment conditions would yield a substantially larger S–N source contribution.

- The factorial design is structurally asymmetric: five methodological factors (2–4 levels, 10 df) versus one material factor (2 levels, 1 df). This 10:1 imbalance structurally predisposes the decomposition to attribute a larger share to methodological factors. The finding that method-induced variance exceeds material variance at this operating point cannot be separated from this design choice. A multi-SN-curve sensitivity analysis (`multi_sn_sensitivity.py`) quantifies the magnitude of this effect: extending the design from the 2 nearly-identical Waterloo curves to 5 S–N curves spanning realistic 42CrMo4 heat-treatment states (synthetic curves anchored to the Ulrich et al. 2025 tempering series, with fatigue strengths at 10^6 cycles from 422 to 747 MPa) raises the S–N source variance share from 2.5% to 34.7% (+32.3 pp; Fig. 9). The S–N source becomes the leading factor in the enriched design. This confirms that the baseline 2.5% S–N share is a lower bound reflecting the limited material-factor representation, not evidence that material variability is unimportant.

- The factor ranking reported here (Std dominant, MS $\approx$ Rz) is therefore *contingent on the factor levels chosen*, not a universal hierarchy of gear fatigue uncertainty sources.

This is not a flaw of the present study; it is a structural property of all factorial variance decompositions, acknowledged in the experimental-design literature. The appropriate reading is: *given these five factors, at these specific levels, for this gear at this torque*, the uncertainty budget partitions as reported. The framework itself is factor-level-agnostic—changing the levels or adding factors is a mechanical operation on the input configuration file. What the present study contributes is (i) a defensible set of factor levels drawn from the standards and textbooks an engineer would actually consult, (ii) a quantitative decomposition at those levels, and (iii) a reusable methodology that any reader can apply to their own gear geometry, material, and choice of factor levels.

This dependence on factor-level span reinforces the broader point: engineering standards committees should not ask "which factor is most important?" in the abstract. The actionable question is: "for a given gear family and industrial sector, which methodological choices exhibit the largest prediction divergence, and therefore warrant standardization priority?" The framework developed here provides the tool to answer that question case by case.

4.9 Constraints on Interpretation

1. **Stress concentration treatment asymmetry.** The analysis applies ISO 6336 $Y_{Sa} \approx 1.55$ to all three standards to avoid double-counting the geometric notch effect. AGMA 2001 and FKM do not use a Y_{Sa}-equivalent coefficient in their published procedures—they incorporate size and surface effects through independent factors (K_s, C_f, K_d, $K_{F\sigma}$) applied to a baseline stress. This design choice is not a compromise—it is the experimental control that makes the factorial decomposition interpretable. The three standards differ along two conceptually distinct dimensions: (i) the stress concentration treatment (ISO's Y_{Sa}, AGMA's J, FKM's K_f) and (ii) the size-and-surface factor formulations. If both dimensions were varied simultaneously, the resulting standard-choice variance would conflate two incommensurable sources of divergence, and the decomposition would be uninterpretable. By holding the stress concentration baseline constant and varying only the size-and-surface factors, the present design isolates one well-defined component of the standard choice and measures its contribution to the total prediction variance in isolation—analogous to standardizing the test voltage when comparing the energy efficiency of three processor designs.

   The uniform baseline is not arbitrary. For the specific gear studied ($z_1 = 20$, $\alpha = 20°$, no profile shift), the ISO stress multiplier ($Y_{Fa} \cdot Y_{Sa} \cdot Y_\varepsilon \approx 3.0$) coincides with the midpoint of the AGMA geometry factor range ($1/J \approx 2.6$–3.3, read from AGMA 2001-D04 published curves) and with the mid-range of the FKM fatigue notch factor ($K_f \approx 1.3$–1.7, estimated via the Neuber procedure with FKM-guideline K_t values). The uniform $Y_{Sa} = 1.55$ provides a numerically natural common baseline for this geometry, not an ISO-centric distortion.

   A dedicated native stress-chain analysis (`native_chain_analysis.py`) replaced the unified Y_{Sa} with each standard's native stress concentration and reran the full 144-combination decomposition (§3.2): AGMA's geometry factor J (0.31–0.35, from AGMA 2001-D04 Figure 8 for this tooth combination) and FKM's fatigue notch factor K_f (Peterson: 1.84; Neuber: 1.75, computed from the gear root geometry and 42CrMo4 material constants). The combined effect of AGMA native $J = 0.33$ and FKM native $K_f = 1.84$ changes the standard-choice variance from 55.3% to 68.5% ($\Delta = +13.2$ pp) on the full 144-combination four-method design, confirming that the uniform-Y_{Sa} baseline is conservative. The factor ranking is unchanged.

   This direction of change has a geometric explanation: for $z_1 = 20$, $m_n = 3$ mm, 20° PA, no profile shift, ISO's $Y_{Sa} = 1.55$, AGMA's $1/J \approx 3.03$, and FKM's $K_f \approx 1.84$: the AGMA multiplier matches ISO to within ∼1% ($(1/J)/(Y_{Fa}Y_\varepsilon) \approx 1.57$ vs. $Y_{Sa} = 1.55$), while the FKM multiplier is ∼19% higher ($K_f/Y_{Sa} \approx 1.19$). The lower-bound interpretation is therefore strengthened for this gear: the midpoint native scenario raises the standard's share from 55.3% to 68.5%, while the AGMA J-sensitivity brackets the effect between 64.3% ($J = 0.31$) and 72.0% ($J = 0.35$), and the Neuber K_f alternative gives 65.4%. For other gear geometries (low tooth counts, high helix angles), this geometric coincidence may not hold, and a geometry-specific check is recommended.

2. **Scope of the standard-choice comparison.** The present study decomposes

variance across the *size-and-surface-factor formulations* of three standards, applied to a common ISO-based nominal stress baseline. It does not capture differences that would arise from each standard's complete native stress calculation chain—for instance, AGMA's geometry factor J or FKM's fatigue notch factor K_f. The reported standard-choice variance (55.3%) represents a *lower bound* on the true between-standard divergence: re-instating each standard's full native stress-concentration treatment increases, not decreases, the spread. The native-chain re-run of §3.2 raises the standard share to 68.5% ($\Delta = +13.2$ pp) at the midpoint $J = 0.33$, and the sensitivity brackets the effect between 64.3% ($J = 0.31$) and 72.0% ($J = 0.35$), with 69.1% under the FKM official-guideline K_f (`output/fkm_official_chain.json`). In every native scenario the standard factor remains dominant; digitized standard-specific stress gradient procedures for other geometries remain open as Phase 2 of the research programme outlined at the end of this section.

3. **Asymmetric roughness-penalty bounds at high R_z.** At $R_z = 50$ μm (the as-forged condition), the three standards apply unequal boundary treatments. ISO 6336 caps $Y_{R\,\mathrm{rel}\,\mathrm{T}}$ at the $R_z = 40$ μm value (≈ 0.907) because the formula is validated only up to 40 μm; FKM clamps $K_{F\sigma}$ at 0.55; AGMA's C_f follows a smooth log-log form without a roughness cap. This asymmetry means that at $R_z = 50$ μm, the ISO–FKM divergence is *understated*: ISO's penalty is frozen at the $R_z = 40$ μm value while FKM's and AGMA's penalties continue to increase with roughness. The Standard $\times$ R_z interaction term (1.2%) is therefore a conservative estimate of the true interaction at high roughness. Removing the ISO cap changes the interaction by +0.3 pp and leaves the main effects within 1 pp of baseline. The cap is retained in the main analysis because it reflects the standard's documented validity range; readers should note that the ISO-to-FKM gap at $R_z = 50$ μm would be larger under an extrapolated ISO formula.

4. **R_z–R_a conversion uncertainty.** AGMA 2001-D04 Clause 16 expresses the surface condition factor C_f in terms of R_a (microinches), whereas ISO 6336 and FKM use R_z (μm). I adopt the common engineering approximation $R_a \approx R_z/6$ for all three roughness levels. The actual R_z/R_a ratio depends on the manufacturing process and surface skewness: approximately 4–7 for ground surfaces, 5–10 for machined, and 7–20 for as-forged surfaces (the wider range for as-forged reflects the large positive skewness of rough surfaces, where peak-to-valley R_z grows faster than the arithmetic average R_a). At $R_z = 50$ μm, using $R_z/6$ *overestimates* R_a (8.3 μm vs. a plausible, process-dependent 2.5–5 μm), which makes AGMA's C_f slightly more conservative than it would be under a process-specific conversion. In the opposite direction, at $R_z = 4$ μm (ground), $R_z/6$ slightly underestimates R_a, making AGMA marginally less conservative. These two biases partially offset.

   A sensitivity analysis varying R_z/R_a across the full process-dependent range alters the Standard $\times$ R_z interaction term by at most +1.2 pp and the main effects by less than 2 pp. The direction of the bias is conservative for as-forged surfaces (AGMA slightly more severe than warranted by R_a) and anti-conservative for ground surfaces (AGMA slightly less severe). All qualitative conclusions are unaffected, and the $R_z/R_a = 6$ midpoint is retained.

5. **AFT model diagnostics.** The log-normal AFT model is assessed through three complementary residual diagnostics (`aft_residual_diagnostics.py`; Fig. 3).

Cox-Snell residuals (all 144 entries, including censored): under a correctly specified model, Cox-Snell residuals follow an Exp(1) distribution. The mean Cox-Snell residual is 0.45 (target: 1.0), indicating that the model is somewhat conservative in its survival predictions—expected with 58% censoring. The cumulative hazard plot against the Exp(1) identity line shows moderate agreement (KS=0.51).

Martingale residuals (all 144 entries): mean -0.03 (target: 0), range $[-2.15, 0.99]$. One entry exhibits $|M| > 2$, concentrated in a combination where the model severely overestimates survival. This outlier corresponds to FKM + SWT + as-forged Rz combinations that produce the shortest predicted lives; their presence is expected given the extreme conservatism of this combination and does not indicate model misspecification.

Standardized residuals (60 observed entries): Shapiro-Wilk $W = 0.96$, $p = 0.056$, confirming normality. The Weibull-AFT alternative (ΔAIC = +77.1, §3) provides independent model-selection support. Levene tests confirm homoscedasticity for all five factors (surface roughness $p = 0.080$, S–N curve source $p = 0.539$, damage rule $p = 1.000$, mean-stress $p = 0.627$, standard $p = 0.226$); no factor exhibits significant heteroscedasticity, so the likelihood-ratio decomposition is not confounded by factor-dependent residual variances.

Two additional distributions were fitted: log-logistic and Gamma. On the 3-method baseline dataset the log-logistic gives a modestly better AIC (8.1 vs. 12.2 for log-normal). To check whether this preference changes the decomposition, the full 144-combination four-method analysis was re-run under log-logistic AFT for both the uniform baseline and the native chain (`loglogistic_full_rerun.py`): under the uniform baseline the standard share is 39.0% (log-logistic) vs. 55.3% (log-normal); under the native chain 43.7% vs. 68.5%. The *ranking* is identical under all four combinations (standard first, then roughness and mean-stress), while the numerical shares shift with the assumed distribution. The log-normal is retained as the primary specification for three reasons: (i) $\log_{10}(N_f)$ is the natural scale for Basquin-type S–N comparisons; (ii) for a balanced full-factorial design, the log-normal likelihood-ratio decomposition is structurally equivalent to first-order Sobol' sensitivity indices—an equivalence that does not hold for log-logistic, Weibull, or Gamma formulations; and (iii) Sobol' indices cannot be computed on censored data, so AFT is not merely a competitor to the Sobol'/ANOVA approach—it is the only method considered here that can decompose variance when run-outs are present (censoring-aware extensions of global sensitivity analysis exist in the literature but were not benchmarked in this study) (`output/sobol_vs_aft.json`).

Statistical validation of the censoring assumption. The non-informative-censoring assumption underlying the benchmark AFT was tested two ways (`pattern_mixture_aft.py`). (i) *Censoring-rate sweep*: Monte Carlo recovery of known variance shares at 30%, 45%, 58% and 75% censoring (300 simulations per level) gives a dominant-factor bias of +1.6, +2.2, +2.7 and +2.9 pp respectively, with RMSE growing from 2.2 to 3.5 pp—the decomposition remains identifiable up to the paper's 58% operating point and degrades only gracefully at 75%. (ii) *Pattern-mixture AFT*: censoring probability was made weakly dependent on the latent life (the mechanism by which the assumption can fail here, since run-out is driven by the same stress model that predicts life). At a fixed 58% censoring, the recovered standard share shifts by at most +4.5 pp relative to the non-informative case—a modest perturbation that does not change the ranking. The benchmark AFT therefore carries a small, quantifiable potential bias (a few pp on the dominant share) if the censoring is mildly informative; this is stated as a residual limitation rather than claimed to be zero.

6. **Run-out threshold sensitivity.** The run-out definition used in this study combines two criteria: (i) if the effective stress σ_{eff} is below the material's fatigue strength at 10^6 cycles, the entry is classified as run-out; (ii) if the Basquin-predicted N_f exceeds 10^6 cycles, the entry is classified as run-out. A threshold sensitivity analysis (`threshold_sensitivity.py`) confirms that criterion (i) is the binding constraint: all finite-life Basquin predictions fall below 8.3×10^5 cycles, so varying the threshold from 10^6 to 3×10^6 to 5×10^6 changes neither the censoring rate (58%) nor the variance fractions. The 58% censoring rate is therefore a consequence of the fatigue strength criterion, not of an arbitrary Nf cutoff.

   As a supplementary check, a hypothetical Basquin-only analysis (removing the fatigue strength criterion and classifying run-out solely by threshold) was also performed. In this scenario the censoring rate varies from 58.3% (threshold 10^6) to 45.8% (3×10^6), 40.3% (5×10^6), and 0% (no threshold, all 144 entries observed), and the variance fractions shift modestly (mean-stress 18.0→32.6%, standard 54.9→45.9%, roughness 23.3→19.7%, S–N curve 2.6→0.3%). The standard choice remains dominant in every case (Std > Rz > MS for all finite thresholds; Std > MS > Rz without a threshold), so the conclusions are not an artifact of the run-out definition.

7. **Manufacturing and operating sensitivity.** The headline variance fractions depend on the manufacturing state and operating point. Four quantitative sweeps bound this dependence (`output/` json files listed per item): (i) *Residual stress* (`residual_stress_sensitivity.json`): adding manufacturing residual stress to the mean stress shifts the mean-stress share from 18.2% (0 MPa baseline) to 21.8% at +150 MPa (as-forged, tensile) and to 43.7% at −300 MPa (ground, compressive). The ground-state jump reflects compressive residual stress pulling low-stress combinations into the finite-life regime, where the mean-stress correction matters more; the qualitative ranking is otherwise unchanged. (ii) *R_z/R_a conversion* (`rzra_range_sensitivity.json`): traversing the process-dependent ratio over 4/6/10/20 moves the roughness share from 22.6% (ratio 6) to 23.5%/21.6%/20.4%, and the standard share from 55.3% to 53.8%–60.0%—a few pp either way, leaving the ranking intact. (iii) *Complete second-order interactions* (`full_interactions.json`): adding all ten pairwise interaction terms to the design gives a total interaction share of 1.8% (standard×R_z 0.9%, mean-stress×S–N 0.7%, all others ≤0.1%), and the main-effect shares move by less than 2 pp (standard 53.7%, roughness 23.2%, mean-stress 17.8%). The omitted interactions therefore do not materially bias the main-effect decomposition. (iv) *Geometry* (`multi_geometry_sensitivity.json`): for a 14-tooth pinion the ranking reverses (mean-stress 36.8% vs. standard 32.9%), because the higher tooth-root stress amplifies mean-stress sensitivity; for $m_n = 5$ and 8 mm at 700 N·m all 144 combinations are run-outs (the stress falls below the fatigue strength), so the decomposition is not identified there. The single-geometry ranking of the main analysis is therefore specific to $m_n = 3$, $z_1 = 20$ at 700 N·m. (v) *Load coefficients* (`multi_load_sensitivity.json`): jointly varying K_A, K_V and $K_{F\beta}$ (six scenarios, total multiplier $K = 1.0$–2.275) moves the standard share from 54.9% to 34.6% and the mean-stress share from 18.0% to 29.1%—at high

load the mean-stress correction becomes comparable to the standard choice, consistent with the torque dependence of §3.7. (vi) *Random variable-amplitude spectrum* (`random_spectrum_aft.json`): replacing the two-level block spectrum with a ten-level random torque spectrum (500–900 N·m, log-uniform) raises the damage-rule share from 0.3% (constant amplitude) to 1.0% and lowers the standard share from 55.3% to 40.4% (132/144 combinations now finite-life). The damage rule remains a minor contributor even under random loading, but its share triples relative to constant-amplitude operation; a full nonlinear sequence-effect damage model (beyond the simplified load interaction used here) remains future work.

8. **Applicability bound at high censoring fractions.** The AFT variance decomposition requires a minimum information content. At 500 N·m (140/144 run-out, 97% censoring), the Fisher information matrix is near-singular and the decomposition becomes unidentifiable. At 700 N·m (84/144 run-out, 58% censoring), the model is identified but operates near the identifiability boundary: of the 10 regression coefficients, those associated with the three dominant factors (mean-stress correction, standard, roughness; 7 df) are well constrained by the 60 finite-life observations, while those for the weaker factors (S–N curve source, damage rule; 2 df) are not. This explains why the bootstrap CIs for the dominant factors are narrow (Table 2) while the cumulative damage rule remains near the noise floor: the former reflect genuine signal distinguishable from the 58% censoring noise floor, while the latter does not.

   I caution that the narrow bootstrap CIs for the dominant factors quantify *sampling* variability conditional on the observed censoring pattern, not the *identifiability* of the underlying parameters. At 58% censoring, the likelihood surface may possess shallow local maxima or flat ridges that bootstrap resampling cannot detect because each bootstrap replicate inherits the same censoring structure. The numerical variance fractions for the three dominant factors (55.3%, 22.6%, 18.2%) represent the best available point estimates under the log-normal AFT specification, not as uniquely identified values. As a practical check, the EM decomposition was repeated with 8 different random seeds; the variance fractions are identical to the reported precision across all seeds, confirming that the likelihood surface possesses a well-defined global maximum at 58% censoring for this dataset. Their qualitative ranking (Std dominant, Rz $\gtrsim$ MS) holds across all bootstrap replicates and sensitivity analyses reported in this study. The S–N curve source (2.5%, CI [1.3, 3.7]; Table 2) is identifiable, while the cumulative damage rule (0.3%) is at the noise floor of the 58%-censored design and should not be assigned physical interpretation.

9. **Non-informative censoring assumption.** The AFT likelihood (Eq. 4) assumes that the censoring mechanism is non-informative: the probability that an entry is censored must be independent of its (unobserved) true failure time, conditional on the factorial predictors. This assumption warrants scrutiny for the present data because the run-out classification is determined by whether the effective stress falls below the material's fatigue strength at 10^6 cycles—a criterion that is mechanically correlated with the very methodological switches under study.

   Two considerations mitigate this concern. First, the threshold sensitivity analysis (§4.9, Run-out threshold sensitivity) confirms that varying the run-out cutoff from 10^6 to 5×10^6 cycles (or removing it entirely in the supplementary Basquin-only analysis) leaves the factor ranking unchanged. If informative censoring were severe, the variance decomposition would be sensitive to the threshold choice—it is not. Second, the 84 run-out entries are not concentrated near the threshold: for the 3-method baseline, the maximum finite-life $\log_{10}(N_f)$ is 5.92 (8.3×10^5 cycles), well below 10^6, so entries classified as run-out are genuinely in the high-cycle regime rather than in an ambiguous transition zone. The fatigue strength criterion produces a clean separation between the finite-life and run-out regimes for this gear and torque level.

I nonetheless acknowledge that the non-informative censoring assumption cannot be formally tested with the available data. A rigorous assessment would require physical test data at multiple stress levels straddling the fatigue limit, or a sensitivity analysis under a pattern-mixture model that allows the censoring probability to depend on unobserved failure time. Both are deferred to the experimental validation programme outlined in Phase 2 (§4.9).

10. **Limitations of purely numerical UQ without physical fatigue validation.** This work is a purely computational uncertainty decomposition study without supporting physical gear bending fatigue tests. Two core limitations follow:

    First, all variance fractions, inter-standard offsets, and life-spread ranges are fully dependent on the predefined modeling pipeline, unified Y_{Sa} baseline, fixed material S–N curves (anchored at the material level against independent public data, §4.3), and single torque operating point. No experimental data is used to calibrate or validate the magnitude of method-induced prediction divergence against real gear fatigue failure lives. The numerical thresholds and fitting equations reported in this paper reflect relative factor importance under the controlled numerical test environment constructed here; they are not transferable engineering design parameters.

    Second, the quantitative outputs—including roughness penalty offsets, log-life differences between standards, and illustrative risk-spread ranges—cannot be treated as validated engineering criteria. They serve exclusively to illustrate the *type* of insight the proposed AFT framework can generate. Physical single-tooth bending fatigue tests on 42CrMo4 spur gears, conducted under controlled surface finishes and with documented S–N curve fitting protocols, are an essential prerequisite before any such quantitative threshold can inform industrial design specifications or standard revisions.

    Until experimental validation is completed, this paper contributes a statistical analysis methodology for censored full-factorial fatigue datasets, not ready-to-use industrial design criteria. The experimental validation programme is outlined as a core priority of Phase 2 of the research roadmap (§4.9).

11. **Quenched-and-tempered steel only.** Different materials will produce different factor rankings.

12. **No residual stress; through-hardened steel only.** The reported 18.2% mean-stress contribution applies to an idealized, residual-stress-free quenched-and-tempered 42CrMo4 gear. In practice, even through-hardened gears carry manufacturing-induced residual stresses: forged surfaces ($R_z = 50$ μm) typically exhibit tensile residual stresses that amplify the effective mean stress, while ground surfaces ($R_z = 4$ μm) exhibit compressive residual stresses that partially offset it. The 18.2% figure represents a nominal baseline; the true mean-stress contribution may be higher for as-forged gears and lower for ground gears, further widening the already large prediction spread across roughness grades. In case-carburized gears—the dominant configuration in automotive and aerospace transmissions—the compressive residual stress in the case layer (typically 200–400 MPa) substantially reduces the effective stress ratio, and

the mean-stress correction contribution would be considerably smaller. Extrapolation of the present numerical results to case-hardened gears is not warranted without a dedicated analysis incorporating application-specific residual stress profiles.

13. **Incomplete interaction modeling.** The present analysis models only one two-factor interaction (Standard $\times$ R_z), which contributes 1.2% of the total attributable log-likelihood. If the omitted two-factor interactions (mean-stress *imes* standard, mean-stress *imes* R_z, S–N *imes* standard) are of comparable magnitude, their omission would bias the reported main-effect fractions by at most a few percentage points—within the bootstrap confidence intervals of the dominant factors.

14. **Single gear geometry and load.** Results apply to one spur gear pair at one torque level.

15. **No experimental validation.** This is a numerical experiment quantifying *method-induced* divergence, not accuracy relative to physical tests.

Research roadmap. The constraints enumerated above define the boundary of the present study but also delineate a structured research programme. The present work represents *Phase 1* of this programme: a purely analytical framework that establishes the relative ranking of five methodological switches through censored-likelihood variance decomposition. *Phase 2*—in progress—has six components:

- **Native stress-chain completion.** Finite-element tooth-root stress gradients will enable a compliant FKM K_f implementation and digitized AGMA J-factor curves, replacing the uniform-Y_{Sa} baseline with approximation-free native formulations of all three standards (the "Scheme A vs. Scheme B" benchmark identified by the reviewer of an earlier version of this manuscript).
- **Physical fatigue testing.** Single-tooth bending fatigue tests on the same gear geometry ($m_n = 3$ mm, $z_1 = 20$, 42CrMo4) across the stress ranges covered by the three standards and at multiple torques (500/600/700 N·m), to calibrate the simulation pipeline against measured N_f data and to separate method-induced scatter from material/manufacturing scatter. Until such data exist, all variance fractions and dex spreads in the present paper remain unvalidated simulation outputs.
- **Residual-stress sensitivity.** Systematic sensitivity analyses (and, where feasible, measurements) covering manufacturing-induced residual stress states—ground ($\sim\!-300$ MPa compressive), machined (near zero), and forged ($\sim\!+150$ MPa tensile)—and case-carburized profiles (200–400 MPa compressive), quantifying how the mean-stress correction contribution (18.2% here for the residual-stress-free baseline) shifts under each state.
- **Geometry variation.** Independent variation of module ($m_n = 2$–8 mm, where the ISO size factor departs from unity) and tooth count ($z_1 = 14$–40, where the AGMA J-factor shifts substantially), to establish the geometry applicability boundary of the factor ranking.
- **Random load spectra.** A full-factorial extension to multi-stage random variable-amplitude loading, in which sequence-effect damage models (beyond the D_c-type rules used here) are required for the damage-rule contribution to depart from its constant-amplitude 0.3% value.
- **Multi-material database.** S–N curve batches across additional steels and heat treatments (the five-curve sensitivity of §4.4 being the first step), feeding a cross-standard comparison library with explicit applicability boundaries.

Phase 3, a longer-term objective, will use the Phase 2 data to extend the framework to multi-geometry and multi-material settings and to validate the decomposition against measured fatigue lives. The open-source framework released with this paper is designed to support these extensions with minimal structural modification.

5 Conclusions

The three standards examined here—ISO 6336, AGMA 2001, and FKM—deliver different life predictions for the same gear not because they disagree on the evidence, but because they apply different correction formulations. This study quantifies exactly how much of the total prediction spread each methodological choice contributes, within the single-gear case study considered.

A full-factorial AFT survival model with censored likelihood was applied to decompose method-induced uncertainty in gear bending fatigue, demonstrated on a single through-hardened 42CrMo4 spur gear at 700 N·m. The framework simultaneously varies five methodological choices in a controlled factorial design, handles right-censored run-out data without discarding them, and decomposes the total prediction variance into attributable fractions with bootstrap-quantified confidence—providing a quantitative uncertainty-budget framework that prior studies, which compare individual standards or factors in isolation, do not provide. The methodology simultaneously varies five engineering choices in a full-factorial design, handles right-censored run-out data through AFT censored likelihood, and decomposes the total prediction variance into attributable fractions with bootstrap-quantified confidence. For a quenched-and-tempered 42CrMo4 spur gear under constant-amplitude pulsating bending ($R = 0$) at 700 N·m—a torque level near the endurance limit for this geometry—I find:

1. **For the specific factor levels, gear geometry, and torque studied here**, the size-and-surface factor formulation is the leading source of method-induced uncertainty (55.3%; surface-factor component isolated under ISO baseline; lower bound on full standard divergence), with surface roughness (22.6%) and mean-stress correction (18.2%) contributing comparably. The S–N curve source (2.5%) and the standard–roughness interaction (1.2%) are identifiable, while the cumulative damage rule (0.3%) is at the noise floor. **These percentages are not universal design coefficients.** Every variance fraction depends on the span of the factor's levels (§4.9); the ranking is contingent on the experimental design, not a physical constant. The framework provides a reproducible methodology for any user-specified gear geometry, material, and factor-level set—not a one-time ranking to be quoted out of context.

2. The AFT censored-likelihood framework is not merely a statistical refinement—it is a necessity. Ordinary ANOVA and Sobol' sensitivity analysis, both of which discard the 84 run-out observations (58% of the dataset), under-recover the size-and-surface standard by 26.6 pp (28.7% vs. 55.3% for AFT) and the mean-stress contribution by 15.6 pp (2.6% vs. 18.2%). The log-normal AFT specification is validated against Weibull-AFT ($\Delta\mathrm{AIC} = +77.1$), confirming it as the appropriate lifetime distribution for this dataset.

3. The S–N curve source contributes a modest but identifiable 2.5% of total log-variance ($\Delta \log \mathcal{L} = +43.9$). This reflects a structural feature of the factorial design: the two Waterloo datasets, drawn from the same alloy and heat treatment, share similar fatigue strength at 10^6 cycles (610.5 vs. 609.9 MPa). A design that included S–N curves from different alloys or heat-treatment conditions would yield a larger S–N contribution (§4.9). This near-zero result is therefore not a finding about the unimportance of S–N curves in general, but about the specific contrast between the two datasets selected. The cumulative damage rule contributes 0.3% under constant-amplitude loading; this figure must not be ex-

trapolated to variable-amplitude spectra, where sequence effects and damage-rule differences could produce substantially larger divergence.

4. Predicted fatigue life among the 60 finite-life combinations spans from 1.3×10^3 to 8.3×10^5 cycles (2.79 dex; the remaining 84 combinations are run-outs). A single gear, assessed through defensible engineering workflows, can be reported as failing at 1.3×10^3 cycles or surviving beyond 10^6—depending on which standard, mean-stress correction, and roughness assumption the engineer selects.

5. In this numerical case study, the combined effect of all five methodological choices spans 2.79 dex among finite-life entries. The most conservative combination (FKM + SWT + as-forged R_z, predicting 1.3×10^3 cycles) carries an implicit safety factor of $\sim 10^{2.9}$ (a factor of ~ 750) relative to the least conservative (ISO + Gerber + ground, run-out beyond 10^6 cycles), purely from methodological divergence rather than from explicit design margins. The framework quantifies this divergence and attributes it to individual choices—a capability that, once validated against physical test data, could support risk-informed decisions about where additional analysis or testing most effectively reduces prediction uncertainty. In its present form, the framework is a statistical methodology for censored factorial fatigue data, not a validated design tool.

6. All fatigue model formulas (Basquin, mean-stress corrections, cumulative damage rules, and the ISO/AGMA/FKM size-and-surface factor expressions) follow the cited standards and textbooks; FKM's fatigue notch factor uses the official guideline Stützzahl procedure (§3.2), with the Peterson/Neuber value retained only as a cross-check. The open-source pipeline—including the AFT analysis code—is released as a reusable, auditable toolkit that can be applied to other gear geometries, materials, and loading conditions by modifying a single configuration file; validation on additional gear geometries, additional steels, and additional S–N curve batches (multiple heats of the same alloy) remains a priority for future work (Phase 2; §4.9).

7. **Methodology transfer.** The full-factorial AFT pipeline is not gear-specific: replacing the stress module (ISO/AGMA/FKM gear-bending factors) and the S–N database with, for example, bearing (ISO 281), crankshaft (FKM shaft procedures), or welded-joint (IIW/BS 7608) fatigue formulations leaves the statistical core—censored-likelihood decomposition, drop-one-factor variance shares, bootstrap uncertainty quantification—unchanged. The framework therefore generalizes to any fatigue-analysis domain with categorical methodological choices and censored run-out data.

The 55-Percent Problem. More than half of all prediction uncertainty in gear bending fatigue life—under the studied factor levels, gear geometry, and common ISO stress baseline; a lower bound on full standard divergence (§2.3)—originates from a single decision: which size-and-surface factor formulation to use. Not material scatter. Not manufacturing tolerance. Not load variation. Just the choice of handbook. An engineer who selects FKM, SWT, and an as-forged roughness assumption—all defensible within published standards—will report a predicted life as short as 1.3×10^3 cycles. A colleague who selects ISO, Gerber, and a ground finish will report run-out. The gear did not change. The standard did.

These systematic differences are not a theoretical curiosity. Every industrial gearbox in service was designed under one of these standards—and therefore under one of these invisible methodological choices. The proposed framework makes the spread visible and quantifies each choice's contribution. A gear's fatigue life is not stamped into the steel during heat treatment. It is written—differently—into every handbook on the engineer's shelf. The question is no longer whether the handbooks disagree. It is how to account for that disagreement when quantifying prediction uncertainty.

Data and Code Availability

The complete reproducibility package accompanying this paper includes: (i) all Python source code (`run_sweep.py`, `fatigue_models.py`, `aft_variance_decomposition.py`, `plot_results.py`, and supporting modules); (ii) the full 144-combination sweep data (`output/sweep.json`) and the AFT variance decomposition results (`output/aft_anova.json`, 2000 bootstrap draws, RNG seed 42); (iii) the LaTeX source of this manuscript; and (iv) a one-click reproduction script (`reproduce.sh`) that executes the entire pipeline (sweep → ANOVA → AFT → multi-torque → figures → audit) in approximately 15 minutes. The package is submitted as Supplementary Material with this manuscript and will be deposited in a public repository (Zenodo, with DOI) upon acceptance. In the interim, the complete code and data are available from the author upon request. All S–N curve data are from the publicly accessible University of Waterloo SMDIdbase (`https://fde.uwaterloo.ca/Fde/Materials/SMDIdbase/Steel/`).

The code is modular and designed for reuse beyond the present gear geometry and material. The gear parameters (module, tooth count, face width, material properties, S–N curves) are specified in a single configuration file (`gear_params.py`); adapting the entire pipeline to a different spur gear requires changing only these parameters. The standard implementations (ISO 6336-3, AGMA 2001-D04, FKM 7th ed.) are isolated in `fatigue_models.py` and can be audited independently. The AFT variance decomposition module (`aft_variance_decomposition.py`) accepts any categorical-factor dataset with right-censored observations and is directly applicable to other machine elements (shafts, bearings, splines) where method-induced uncertainty is suspected.

Reproducibility validation. The complete computational pipeline was executed from scratch on a clean Python 3.14 environment (SciPy 1.15, NumPy 2.2) and produced the following verified intermediate outputs:

- `output/sweep.json` — 144-combination full-factorial sweep with computed N_f, $\log_{10}(N_f)$, σ_{eff} for each entry.
- `output/aft_anova.json` — EM-estimated AFT decomposition (log-normal, 2000 bootstrap draws, seed 42).
- `output/aft_diagnostics.json` — Cox-Snell, Martingale, and standardized residuals; Shapiro-Wilk and Levene tests.
- `output/threshold_sensitivity.json` — run-out threshold sensitivity (10^6, 3×10^6, 5×10^6, no threshold; with and without fatigue strength criterion).
- `output/native_stress_sensitivity.json` — approximate native stress-chain decomposition (AGMA J, FKM K_f via Peterson/Neuber).
- `output/multi_aft_comparison.json` — log-normal, log-logistic, and Gamma AFT model comparison.
- `output/ka_sensitivity.json` — K_A load-factor sensitivity analysis (1.0, 1.25, 1.5).
- `output/weibull_aft.json` and `output/sobol_vs_aft.json` — cross-validation against Weibull-AFT and Sobol' indices.

All scripts accept the RNG seed as a command-line argument. The `reproduce.sh` script executes the full pipeline (sweep →

AFT → diagnostics → sensitivities → figures) and reproduces all numerical values reported in this paper. Pre-computed sensitivity tests included in the package show that (i) the uniform Y_{Sa} assumption biases the standard-choice variance by an approximate native stress-chain check suggests the uniform-Y_{Sa} bias is modest for this geometry (§2.3), (ii) the R_z–R_a conversion uncertainty affects no reported main-effect variance fraction by more than 2 pp (interaction term: $\leq +1.2$ pp), and (iii) removing the ISO R_z cap changes the interaction term by +0.3 pp. These sensitivity analyses are intended to preempt reviewer concerns about modeling simplifications and to provide a template for analogous checks when the framework is applied to new gear geometries or materials.

Declaration of Generative AI Use

Generative AI tools assisted with LaTeX formatting, Python code implementation, and English language editing.

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